NOISY ℓ^0-SPARSE SUBSPACE CLUSTERING ON DIMENSIONALITY REDUCED DATA

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ABSTRACT

High-dimensional data often lie in or close to low-dimensional subspaces. Sparse subspace clustering methods with sparsity induced by ℓ^0-norm, such as ℓ^0-Sparse Subspace Clustering (ℓ^0-SSC) (Yang et al., 2016), are demonstrated to be more effective than its ℓ^1 counterpart such as Sparse Subspace Clustering (SSC) (Elhamifar & Vidal, 2013). However, the theoretical analysis of ℓ^0-SSC is restricted to clean data that lie exactly in subspaces. Real data often suffer from noise and they may lie close to subspaces. In this paper, we show that an optimal solution to the optimization problem of noisy ℓ^0-SSC achieves Subspace Detection Property (SDP), a key element with which data from different subspaces are separated, under deterministic and randomized models. Our results provide theoretical guarantee on the correctness of noisy ℓ^0-SSC in terms of SDP on noisy data for the first time. In order to improve the efficiency of noisy ℓ^0-SSC, we propose Noisy-DR-ℓ^0-SSC and Noisy-DR-ℓ^0-SSC-OSNAP which provably recover the subspaces on dimensionality reduced data. Both algorithms first project the data onto a lower dimensional space by linear transformation, then perform noisy ℓ^0-SSC on the dimensionality reduced data for improved efficiency. Experimental results demonstrate the effectiveness of Noisy-DR-ℓ^0-SSC and Noisy-DR-ℓ^0-SSC-OSNAP.

1 INTRODUCTION

Clustering is an important unsupervised learning procedure for analyzing a broad class of scientific data. High-dimensional data, such as facial images and gene expression data, often lie in low-dimensional subspaces in many cases, and clustering in accordance to the underlying subspace structure is particularly important. For example, the well-known Principal Component Analysis (PCA) works perfectly if the data are distributed around a single subspace. The subspace learning literature develops more general methods that recover multiple subspaces in the original data, and subspace clustering algorithms (Vidal, 2011) aim to partition the data such that data belonging to the same subspace are identified as one cluster. Among various subspace clustering algorithms, the ones that employ sparsity prior, such as Sparse Subspace Clustering (SSC) (Elhamifar & Vidal, 2013) and ℓ^0-Sparse Subspace Clustering (ℓ^0-SSC) (Yang et al., 2016), have been proven to be effective in separating the data in accordance with the subspaces that the data lie in under certain assumptions.

Sparse subspace clustering methods construct the sparse similarity matrix by sparse representation of the data. Subspace detection property (SDP) defined in Section 2.2 ensures that the similarity between data from different subspaces vanishes in the sparse similarity matrix, and applying spectral clustering (Ng et al., 2001) on such sparse similarity matrix leads to compelling clustering performance. Elhamifar and Vidal (Elhamifar & Vidal, 2013) prove that when the subspaces are independent or disjoint, SDP can be satisfied by solving the canonical sparse linear representation problem using data as the dictionary, under certain conditions on the rank, or singular value of the data matrix and the principle angle between the subspaces. SSC has been successfully applied to a novel deep neural network architecture, leading to the first deep sparse subspace clustering method (Peng et al., 2016). Under the independence assumption on the subspaces, low rank representation (Liu et al., 2010; 2013) is also proposed to recover the subspace structures. Relaxing the assumptions on the subspaces to allowing overlapping subspaces, the Greedy Subspace Clustering (Park et al., 2014) and the Low-Rank Sparse Subspace Clustering (Wang et al., 2013) achieve subspace detection property with high probability. The geometric analysis in (Soltanolkotabi & Candés, 2012)

shows the theoretical results on subspace recovery by SSC. In the following text, we use the term SSC or ℓ^1-SSC exchangeably to indicate the Sparse Subspace Clustering method in (Elhamifar & Vidal, 2013).

Real data often suffer from noise. The correctness of noisy SSC is analyzed in (Wang & Xu, 2013) which handles noisy data that lie close to disjoint or overlapping subspaces, and the original optimization problem of noisy SSC is proposed in (Elhamifar & Vidal, 2013). While ℓ^0-SSC (Yang et al., 2016) renders better empirical performance than ℓ^1-SSC, its correctness is only proved for clean data based on a constrained ℓ^0-minimization problem, and ℓ^0-SSC empirically solves an unconstrained ℓ^0-regularized problem to handle noise in data. As a result, it lacks theoretical guarantee of the correctness on noisy data. In the sequel, we refer to ℓ^0-SSC for noisy data with the unconstrained ℓ^0-regularized problem in (Yang et al., 2016) as noisy ℓ^0-SSC. In this paper, the correctness of noisy ℓ^0-SSC on noisy data in terms of the subspace detection property is established. Our analysis is under both deterministic model and randomized models, which are also the models employed in the geometric analysis of SSC (Soltanolkotabi & Candés, 2012). Our randomized analysis demonstrates potential advantage of noisy ℓ^0-SSC over its ℓ^1 counterpart as more general assumption on data distribution can be adopted. Due to the high dimension of the original data, the computational cost of noisy ℓ^0-SSC is also high. To address this problem, we present Noisy Dimensionality Reduced ℓ^0-Sparse Subspace Clustering (Noisy-DR-ℓ^0-SSC) and Noisy-DR-ℓ^0-SSC-OSNAP, which improve the efficiency of noisy ℓ^0-SSC while provably enjoying theoretical correctness and robustness to noise.

1.1 Contributions

Firstly, we propose two novel algorithms improving the efficiency of noisy ℓ^0-SSC, Noisy Dimensionality Reduced ℓ^0-Sparse Subspace Clustering (Noisy-DR-ℓ^0-SSC) and Noisy-DR-ℓ^0-SSC-OSNAP in Section 3. Both algorithms accelerate noisy ℓ^0-SSC and they enjoy correctness in terms of the subspace detection property under the deterministic model for noisy data. Noisy-DR-ℓ^0-SSC first projects the data onto a lower dimensional space by random projection, then performs noisy ℓ^0-SSC on the dimensionality reduced data. Noisy-DR-ℓ^0-SSC-OSNAP further improves the efficiency of Noisy-DR-ℓ^0-SSC by using the sparse linear transformation according to Oblivious Sparse Norm-Approximating Projection (OSNAP) (Nelson & Nguyen, 2013). Experimental results demonstrate the effectiveness of both Noisy-DR-ℓ^0-SSC and Noisy-DR-ℓ^0-SSC-OSNAP.

Secondly, the correctness of noisy ℓ^0-SSC on noisy data in terms of the subspace detection property is established for the first time, which is presented in Section A in the appendix of this paper. Our analysis is under both deterministic model and randomized models, which are also the models employed by the geometric analysis of SSC (Soltanolkotabi & Candés, 2012). Our randomized analysis demonstrates potential advantage of noisy ℓ^0-SSC over its ℓ^1 counterpart as noisy ℓ^0-SSC requires less restrictive assumptions. The analysis of noisy ℓ^0-SSC lays the foundation for our analysis of Noisy-DR-ℓ^0-SSC and Noisy-DR-ℓ^0-SSC-OSNAP.

1.2 Notations

We use bold letters for matrices and vectors, and regular lower letter for scalars throughout this paper. The bold letter with superscript indicates the corresponding column of a matrix, e.g. $\mathbf{A}^i$ is the i-th column of matrix $\mathbf{A}$, and the bold letter with subscript indicates the corresponding element of a matrix or vector. $\|\cdot\|_F$ and $\|\cdot\|_p$ denote the Frobenius norm and the vector ℓ^p-norm or the matrix p-norm, and $\|\cdot\|_0$ is the ℓ^0-norm, that is, the number of nonzero elements of a vector. $\mathrm{diag}(\cdot)$ indicates the diagonal elements of a matrix. $\mathbf{H_T} \subseteq \mathbb{R}^d$ indicates the subspace spanned by the columns of $\mathbf{T}$, and $\mathbf{A_I}$ denotes a submatrix of $\mathbf{A}$ whose columns correspond to the nonzero elements of $\mathbf{I}$ (or with indices in $\mathbf{I}$ without confusion). $\sigma_t(\cdot)$ denotes the t-th largest singular value of a matrix, and $\sigma_{\min}(\cdot)$ indicates the smallest singular value of a matrix. $\mathrm{supp}(\cdot)$ is the support of a vector, $\mathbb{P}_{\mathcal{S}'}$ is the operator of orthogonal projection onto the subspace $\mathcal{S}'$. $[n]$ represents all the natural numbers between 1 and n inclusively.

2 PROBLEM SETUP

2.1 NOTATIONS

We hereby introduce the notations for subspace clustering on noisy data considered in this paper. The uncorrupted data matrix is denoted by $\mathbf{Y} = [\mathbf{y}_1, \ldots, \mathbf{y}_n] \in \mathbb{R}^{d\times n}$, where d is the dimensionality and n is the size of the data. The uncorrupted data $\mathbf{Y}$ lie in a union of K distinct subspaces $\{\mathcal{S}_k\}_{k=1}^K$ of dimensions $\{d_k\}_{k=1}^K$. The observed noisy data is $\mathbf{X} = \mathbf{Y} + \mathbf{N}$, where $\mathbf{N} = [\mathbf{n}_1, \ldots, \mathbf{n}_n] \in \mathbb{R}^{d\times n}$ is the additive noise. $\mathbf{x}_i = \mathbf{y}_i + \mathbf{n}_i$ is the noisy data point that is corrupted by the noise $\mathbf{n}_i$. Let $\mathbf{Y}^{(k)} \in \mathbb{R}^{d\times n_k}$ denote the data belonging to subspace $\mathcal{S}_k$ with $\sum_{k=1}^K n_k = n$, and denote the corresponding columns in $\mathbf{X}$ by $\mathbf{X}^{(k)}$. The data $\mathbf{X}$ are normalized such that each column has unit ℓ^2-norm in our deterministic analysis. We consider deterministic noise model where the noise $\mathbf{Z}$ is fixed and $\max \|\mathbf{n}_i\| \leq \delta$. Note that our analysis can be extended to a random noise model which is common and also considered by noisy SSC (Wang & Xu, 2013), and the random noise model assumes that columns of $\mathbf{Z}$ are sampled i.i.d. and $\max \|\mathbf{n}_i\| \leq \delta$ with high probability. Note that such random noise model does not require spherical symmetric noise as that in (Wang & Xu, 2013).

2.2 REVISIT ℓ^0-SSC

ℓ^0-SSC (Yang et al., 2016) proposes to solve the following ℓ^0 sparse representation problem

$$\min_{\mathbf{Z}} \|\mathbf{Z}\|_0 \quad s.t.\ \mathbf{X} = \mathbf{X}\mathbf{Z},\ \operatorname{diag}(\mathbf{Z}) = \mathbf{0}, \tag{1}$$

and it proves that the subspace detection property defined in Definition 2.1 is satisfied with the globally optimal solution to (1). In (Yang et al., 2016), the ℓ^0 regularized sparse approximation problem below is solved so as to handle noisy data for ℓ^0-SSC, which is the optimization problem of noisy ℓ^0-SSC:

$$\min_{\mathbf{Z}\in\mathbb{R}^{n\times n}, \operatorname{diag}(\mathbf{Z})=\mathbf{0}} L(\mathbf{Z}) = \|\mathbf{X} - \mathbf{X}\mathbf{Z}\|_F^2 + \lambda\|\mathbf{Z}\|_0. \tag{2}$$

The optimization problem of noisy ℓ^0-SSC (2) is separable. For each $1 \leq i \leq n$, the optimization problem with respect to the sparse code of i-th data point is

$$\min_{\boldsymbol{\beta}\in\mathbb{R}^n, \boldsymbol{\beta}_i=0} L(\boldsymbol{\beta}) = \|\mathbf{x}_i - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \lambda\|\boldsymbol{\beta}\|_0. \tag{3}$$

The definition of subspace detection property for noisy ℓ^0-SSC and noiseless ℓ^0-SSC, i.e. ℓ^0-SSC on noiseless data, is defined in Definition 2.1 below.

Definition 2.1. (Subspace detection property for noisy and noiseless ℓ^0-SSC) Let $\mathbf{Z}^*$ be an optimal solution to (2). The subspaces $\{\mathcal{S}_k\}_{k=1}^K$ and the data $\mathbf{X}$ satisfy the Subspace Detection Property (SDP) for noisy ℓ^0-SSC if $\mathbf{Z}^i$ is a nonzero vector, and nonzero elements of $\mathbf{Z}^i$ correspond to the columns of $\mathbf{X}$ from the same subspace as $\mathbf{y}_i$ for all $1 \leq i \leq n$. Similarly, in the noiseless setting where $\mathbf{X} = \mathbf{Y}$, let $\mathbf{Z}^*$ be an optimal solution to (1). The subspaces $\{\mathcal{S}_k\}_{k=1}^K$ and the data $\mathbf{X}$ satisfy the SDP for noiseless ℓ^0-SSC if $\mathbf{Z}^i$ is a nonzero vector, and nonzero elements of $\mathbf{Z}^i$ correspond to the columns of $\mathbf{X}$ that from the same subspace as $\mathbf{y}_i$ for all $1 \leq i \leq n$. We say that SDP holds for $\mathbf{x}_i$ if nonzero elements of $\mathbf{Z}^{*i}$ correspond to the data that lie in the same subspace as $\mathbf{y}_i$, for either noisy ℓ^0-SSC or noiseless ℓ^0-SSC.

It should be emphasized that the theoretical analysis of ℓ^0-SSC (Yang et al., 2016) is limited to clean data, as a result, it is not clear if SDP can hold for an optimal solution to the noisy ℓ^0-SSC problem. In Section A of the appendix, we prove the correctness of noisy ℓ^0-SSC in terms of SDP under both deterministic and randomized models for this first time. Under the deterministic model, the subspaces and the data in each subspace are fixed. Please refer to Section A.1 of the appendix for the details of all the details of deterministic and randomized models.

Before presenting our methods of noisy ℓ^0-SSC on dimensionality reduced data, we introduce the definitions of general positions and external subspace which are crucial for the theoretical analysis.

Definition 2.2. (General position) For any $1 \leq k \leq K$, the data $\mathbf{Y}^{(k)}$ are in general position if any subset of $L \leq d_k$ data points (columns) of $\mathbf{Y}^{(k)}$ are linearly independent. $\mathbf{Y}$ are in general position if $\mathbf{Y}^{(k)}$ are in general position for $1 \leq k \leq K$.

The assumption of general condition is rather mild. In fact, if the data points in $\mathbf{X}^{(k)}$ are independently distributed according to any continuous distribution, then they almost surely in general position.

Let the distance between a point $\mathbf{x} \in \mathbb{R}^d$ and a subspace $\mathcal{S} \subseteq \mathbb{R}^d$ be defined as $d(\mathbf{x}, \mathcal{S}) = \inf_{\mathbf{y} \in \mathcal{S}} \|\mathbf{x} - \mathbf{y}\|_2$, the definition of external subspaces is presented as follows. Figure 2 in the appendix illustrates an example of external subspace.

Definition 2.3. (External subspace) For a point $\mathbf{y} \in \mathbf{Y}^{(k)}$, a subspace $\mathbf{H}_{\{\mathbf{y}_{i_j}\}_{j=1}^L}$ spanned by a set of linear independent points $\{\mathbf{y}_{i_j}\}_{j=1}^L \subseteq \mathbf{Y}$ is defined to be an external subspace of $\mathbf{y}$ if $\{\mathbf{y}_{i_j}\}_{j=1}^L \nsubseteq \mathbf{Y}^{(k)}$ and $\mathbf{y} \notin \{\mathbf{y}_{i_j}\}_{j=1}^L$. The point $\mathbf{y}$ is said to be away from its external subspaces if $\min_{\mathbf{H} \in \mathcal{H}_{\mathbf{y},d_k}} d(\mathbf{y}, \mathbf{H}) > 0$, where $\mathcal{H}_{\mathbf{y},d}$ are the set of all external subspaces of $\mathbf{y}$ of dimension no greater than d for $\mathbf{y}$, i.e. $\mathcal{H}_{\mathbf{y},d} = \{\mathbf{H} \colon \mathbf{H} = \mathbf{H}_{\{\mathbf{y}_{i_j}\}_{j=1}^L}, \dim[\mathbf{H}] = L, L \leq d, \{\mathbf{y}_{i_j}\}_{j=1}^L \nsubseteq \mathbf{Y}^{(k)}, \mathbf{y} \notin \{\mathbf{y}_{i_j}\}_{j=1}^L\}$. All the data points in $\mathbf{Y}^{(k)}$ are said to be away from the external subspaces if each of them is away from the its associated external spaces.

3 Noisy ℓ^0-SSC on Dimensionality Reduced Data

Albeit the theoretical guarantee and compelling empirical performance of noisy ℓ^0-SSC, the computational cost of noisy ℓ^0-SSC is high with the high dimensionality of the data. In this section, we propose two methods of performing noisy ℓ^0-SSC on dimensionality reduced data, which are Noisy-DR-ℓ^0-SSC and Noisy-DR-ℓ^0-SSC-OSNAP, so as to improve the efficiency of noisy ℓ^0-SSC. Both methods apply a linear transformation to the original data so as to reduce its dimension, then perform noisy ℓ^0-SSC on the dimensionality reduced data. Noisy-DR-ℓ^0-SSC uses the linear transformation induced by randomized low-rank approximation of the original data. Noisy-DR-ℓ^0-SSC-OSNAP is even more efficient than Noisy-DR-ℓ^0-SSC by using the sparse linear transformation according to Oblivious Sparse Norm-Approximating Projection (OSNAP) (Nelson & Nguyen, 2013). The theoretical guarantee on the correctness of Noisy-DR-ℓ^0-SSC and Noisy-DR-ℓ^0-SSC-OSNAP under deterministic models is presented.

3.1 Method

Noisy-DR-ℓ^0-SSC performs subspace clustering by the following two steps: 1) obtain the dimension reduced data $\tilde{\mathbf{X}} = \mathbf{P}\mathbf{X}$ with a linear transformation $\mathbf{P} \in \mathbb{R}^{p \times d}$ ($p < d$). 2) perform noisy ℓ^0-SSC on the compressed data $\tilde{\mathbf{X}}$:

$$\min_{\tilde{\boldsymbol{\beta}} \in \mathbb{R}^n, \tilde{\boldsymbol{\beta}}_i = 0} L(\tilde{\boldsymbol{\beta}}) = \|\tilde{\mathbf{x}}_i - \tilde{\mathbf{X}}\tilde{\boldsymbol{\beta}}\|_2^2 + \tilde{\lambda}\|\tilde{\boldsymbol{\beta}}\|_0. \tag{4}$$

If $p < d$, Noisy-DR-ℓ^0-SSC operates on the compressed data $\tilde{\mathbf{X}}$ rather than on the original data, so that its efficiency is improved.

3.2 Analysis

High-dimensional data often exhibits low-dimensional structures, which often leads to low-rankness of the data matrix. Intuitively, if the data is low rank, then it could be safe to perform noisy ℓ^0-SSC on its dimensionality reduced version by the linear projection $\mathbf{P}$, and it is expected that $\mathbf{P}$ can preserve the information of the subspaces contained in the original data as much as possible, while effectively removing uninformative dimensions.

To this end, we propose to choose $\mathbf{P}$ as a random projection induced by randomized low-rank approximation of the data in this subsection. The key idea is to obtain an approximate low-rank decomposition of the data. Using the random projection induced by such low-rank approximation

as the linear transformation $\mathbf{P}$, the clustering correctness still holds for Noisy-DR-ℓ^0-SSC with a high probability.

Randomized algorithms are efficient and they have been extensively studied in the computer science and numerical linear algebra literature. They have been employed to accelerate various numerical matrix computation and matrix optimization problems, including random projection for matrix decomposition (Frieze et al., 2004; Drineas et al., 2004; Sarlos, 2006; Drineas et al., 2006; 2008; Mahoney & Drineas, 2009; Drineas et al., 2011; Lu et al., 2013).

Formally, a random matrix $\mathbf{T} \in \mathbb{R}^{n\times p}$ is generated such that each element $\mathbf{T}_{ij}$ is sampled independently according to the Gaussian distribution $\mathcal{N}(0,1)$. QR decomposition is then performed on $\mathbf{XT}$ to obtain the basis of its column space, namely $\mathbf{XT} = \mathbf{QR}$ where $\mathbf{Q} \in \mathbb{R}^{d\times p}$ is an orthogonal matrix of rank p and $\mathbf{R} \in \mathbb{R}^{p\times p}$ is an upper triangle matrix. The columns of $\mathbf{Q}$ form the orthogonal basis for the sample matrix $\mathbf{XT}$. An approximation of $\mathbf{X}$ is then obtained by projecting $\mathbf{X}$ onto the column space of $\mathbf{XT}$: $\mathbf{QQ}^\top\mathbf{X} = \mathbf{QW} = \widehat{\mathbf{X}}$ where $\mathbf{W} = \mathbf{Q}^\top\mathbf{X} \in \mathbb{R}^{p\times n}$. In this manner, a randomized low-rank decomposition of $\mathbf{X}$ is achieved by $\widehat{\mathbf{X}} = \mathbf{QW}$.

We present probabilistic result on the correctness of Noisy-DR-ℓ^0-SSC using the random projection induced by randomized low-rank decomposition of the data $\mathbf{X}$, namely $\mathbf{P} = \mathbf{Q}^\top$. In the sequel, $\tilde{\mathbf{x}} = \mathbf{P}\mathbf{x}$ for any $\mathbf{x} \in \mathbb{R}^n$. Thanks to the analysis of noisy ℓ^0-SSC, in order to guarantee the subspace detection property on the dimensionality-reduced data $\tilde{\mathbf{X}}$, it is crucial to make sure that the conditions, such as (29) and (30) in Theorem A.4, of Section A still hold after the linear transformation.

We denote by $\tilde{\boldsymbol{\beta}}^*$ an optimal solution to (4). We also define the following quantities in the analysis of the subspace detection property, which correspond to M_i, $\bar{\sigma}_{\mathbf{Y},r}$, $\sigma_{\mathbf{X},r}$ and μ_r used in the analysis on the original data:

$$\tilde{M}_i \triangleq \min\{d(\tilde{\mathbf{y}}_i, \mathbf{H})\colon \mathbf{H} \in \mathcal{H}_{\tilde{\mathbf{y}}_i,\tilde{d}_k}\}, \tag{5}$$

where $\mathcal{H}_{\tilde{\mathbf{y}}_i,\tilde{d}_k}$ is all the external subspaces of $\tilde{\mathbf{y}}_i$ with dimension no greater than $\tilde{d}_k$ in the transformed space by $\mathbf{P}$.

$$\bar{\sigma}_{\tilde{\mathbf{Y}},r} \triangleq \min_{\boldsymbol{\beta}:\|\boldsymbol{\beta}\|_0=r,\mathrm{rank}(\tilde{\mathbf{Y}}_{\boldsymbol{\beta}})=\|\boldsymbol{\beta}\|_0} \sigma_{\min}(\tilde{\mathbf{Y}}_{\boldsymbol{\beta}}), \tag{6}$$

$$\sigma_{\tilde{\mathbf{X}},r} \triangleq \min\{\sigma_{\min}(\tilde{\mathbf{X}}_{\boldsymbol{\beta}})\colon 1 \le \|\boldsymbol{\beta}\|_0 \le r\}, \tag{7}$$

$$\tilde{\mu}_r \triangleq \frac{\delta}{\min_{1\le r'<r} \bar{\sigma}_{\tilde{\mathbf{Y}},r} - \delta}. \tag{8}$$

Theorem 3.1. (Subspace detection property holds for Noisy-DR-ℓ^0-SSC under deterministic model) Let nonzero vector $\boldsymbol{\beta}^*$ be an optimal solution to the noisy ℓ^0-SSC problem (3) for point $\mathbf{x}_i$ with $\|\boldsymbol{\beta}^*\|_0 = r^*$, $n_k \ge d_k + 1$ for every $1 \le k \le K$, and there exists $1 < r_0 \le d$ such that $1 < r^* \le r_0$. Suppose $\mathbf{Y}$ is in general position, $\delta < \min_{1\le r<r_0} \bar{\sigma}_{\mathbf{Y},r}$, and $\tilde{M}_{i,\delta} \triangleq \tilde{M}_i - \delta$. Suppose the following conditions hold:

(i)
$$C_{p,p_0} + 2\delta\sqrt{\tilde{d}_{\max}} < \min_{k=1,\ldots,K} \sigma_{\mathbf{Y}}^{(k)}, \tag{9}$$
where $\tilde{d}_{\max} \triangleq \max_k \tilde{d}_k$, $\sigma_{\mathbf{Y}}^{(k)} \triangleq \min\{\sigma_{\min}(\mathbf{A})\colon \mathbf{A} \subseteq \mathbf{Y}^{(k)}, \mathbf{A} \in \mathbb{R}^{d\times n'}, n' \le \tilde{d}_k\}$.

(ii) $\delta(1 + 2\sqrt{r_0}) < \min_{1\le r<r_0} \bar{\sigma}_{\mathbf{Y},r} - C_{p,p_0}$,

(iii) $\min_{1\le r\le \tilde{d}_k} \sigma_{\mathbf{Y},r} > C_{p,p_0} - 2\delta\sqrt{\tilde{d}_k}$ and
$$M_i - C_{p,p_0}\Big(1 + \frac{1}{\min_{1\le r\le \tilde{d}_k} \sigma_{\mathbf{Y},r} - C_{p,p_0} - 2\delta\sqrt{\tilde{d}_k}}\Big) > \delta + \frac{2\delta}{\sigma_{\mathbf{X},r_0} - C_{p,p_0}}, \tag{10}$$
for all $\mathbf{y}_i \in \mathcal{S}_k$ and $1 \le k \le K$.

(iv) $\min_{1\leq r<r_0}\bar{\sigma}_{\mathbf{Y},r_0} > C_{p,p_0} - 2\delta\sqrt{r_0} - \delta$ and

$$\frac{\delta}{\min_{1\leq r<r_0}\bar{\sigma}_{\mathbf{Y},r_0} - C_{p,p_0} - 2\delta\sqrt{r_0} - \delta} < 1 - \frac{2\delta}{\sigma_{\mathbf{X},r_0} - C_{p,p_0}}. \tag{11}$$

If

$$\tilde{\lambda}_0 < \tilde{\lambda} < 1, \tag{12}$$

where $\tilde{\lambda}_0 = \max\{\max\{\tilde{\lambda}_1, \tilde{\lambda}_2, \frac{1}{r_0}\}$ and

$$\tilde{\lambda}_1 = \inf\{0 < \tilde{\lambda} < 1 \colon \sqrt{1-\tilde{\lambda}} + \frac{2\delta}{\sigma_{\tilde{\mathbf{X}},r_0}\sqrt{\tilde{\lambda}}} < \tilde{M}_{i,\delta}\}, \tag{13}$$

$$\tilde{\lambda}_2 = \inf\{0 < \tilde{\lambda} < 1 \colon \tilde{\lambda} - \frac{2\delta}{\sigma_{\mathbf{X},r_0}}\frac{1}{\sqrt{\tilde{\lambda}}} > \tilde{\mu}_{r_0}\}, \tag{14}$$

then with probability at least $1 - 6e^{-p}$, the subspace detection property holds for $\tilde{\mathbf{x}}_i$ with $\tilde{\boldsymbol{\beta}}^*$. Here $\tilde{M}_i$, $\tilde{\mu}_r$ and $\tilde{\sigma}_{\tilde{\mathbf{X}},r_0}$ are defined in (63), 66) and (65) respectively.

3.3 The Algorithm of Noisy-DR-ℓ^0-SSC

Algorithm 2 describes how to perform Noisy-DR-ℓ^0-SSC for data clustering. Note that Noisy-DR-ℓ^0-SSC performs noisy ℓ^0-SSC on the dimensionality reduced data $\tilde{\mathbf{X}}$. Proximal Gradient Descent (PGD) is employed to optimize the objective function of noisy ℓ^0-SSC for every data point $\mathbf{x}_i$, which is described in Algorithm 1. In the t-th iteration of PGD for problem (3), the variable $\boldsymbol{\beta}$ is updated according to

$$\boldsymbol{\beta}^{(t+1)} = T_{\sqrt{2\lambda s}}(\boldsymbol{\beta}^{(t)} - s\nabla g(\boldsymbol{\beta}^{(t)})), \tag{15}$$

where s is a positive step size, $g(\boldsymbol{\beta}) = \|\mathbf{x}_i - \mathbf{X}\boldsymbol{\beta}\|_2^2$, T_θ is an element-wise hard thresholding operator:

$$[T_\theta(\mathbf{u})]_j = \begin{cases} 0 & : \ |\mathbf{u}_j| \leq \theta \\ \mathbf{u}_j & : \ \text{otherwise} \end{cases}, \quad 1 \leq j \leq n.$$

It is proved in (Yang et al., 2017) that the sequence $\{\boldsymbol{\beta}^{(t)}\}$ generated by PGD converges to a critical point of (3).

Algorithm 1 Proximal Gradient Descent (PGD) for noisy ℓ^0-SSC problem (3)

1: Input: the initialization $\boldsymbol{\beta}^{(0)}$, step size $s > 0$, parameter λ, maximum iteration number T, stopping threshold ε
2: **for** $1 \leq t \leq T$ **do**
3: $\quad\tilde{\boldsymbol{\beta}}^{(t)} = \boldsymbol{\beta}^{(t-1)} - s\nabla g(\boldsymbol{\beta}^{(t-1)})$
4: $\quad\boldsymbol{\beta}^{(t)} = T_{\sqrt{2\lambda s}}(\tilde{\boldsymbol{\beta}}^{(t)})$, $\boldsymbol{\beta}_i^{(t)} = 0$
5: $\quad$**if** $|L(\boldsymbol{\beta}^{(t)}) - L(\boldsymbol{\beta}^{(t-1)})| < \varepsilon$ **then**
6: $\quad\quad$**break**
7: $\quad$**end if**
8: **end for**
9: Output: $\widehat{\boldsymbol{\beta}}$ which is the suboptimal solution to (3)

Algorithm 2 Noisy Dimensionality Reduced ℓ^0-Sparse Subspace Clustering (Noisy-DR-ℓ^0-SSC)

1: Generate a Gaussian random matrix $\mathbf{T} \in \mathbb{R}^{n\times p}$ where each element $\mathbf{T}_{ij}$ is sampled independently according to the standard Gaussian distribution $\mathcal{N}(0,1)$
2: Perform QR decomposition on $\mathbf{XT}$, $\mathbf{XT} = \mathbf{QR}$ where $\mathbf{Q} \in \mathbb{R}^{d\times p}$
3: Set the linear transformation $\mathbf{P} = \mathbf{Q}^\top$, and obtain the dimensionality reduced data $\tilde{\mathbf{X}} = \mathbf{PX}$
4: Perform noisy ℓ^0-SSC on $\tilde{\mathbf{X}}$ using Algorithm 1

We further study the case when the linear transformation $\mathbf{P}$ for Noisy-DR-ℓ^0-SSC is a sparse matrix constructed by Oblivious Sparse Norm-Approximating Projection (OSNAP) (Nelson & Nguyen, 2013), so that the efficiency is further improved. The resultant algorithm is termed Noisy-DR-ℓ^0-SSC-OSNAP whose theoretical property is demonstrated in the next subsection.

3.4 NOISY-DR-ℓ^0-SSC-OSNAP

In this subsection, we study Noisy-DR-ℓ^0-SSC-OSNAP, where the linear transformation $\mathbf{P}$ for Noisy-DR-ℓ^0-SSC is replaced by a sparse matrix according to the Oblivious Sparse Norm-Approximating Projection (OSNAP) (Nelson & Nguyen, 2013). We choose $\mathbf{P}$ as one the most efficient version of OSNAP, where each column of $\mathbf{P}$ only has 1 nonzero element. Each column of $\mathbf{P}$ is chosen to have exactly one nonzero element in a random location, and each nonzero element is ± 1 uniformly at random. Such $\mathbf{P}$ enjoys ℓ^2-norm preserving property stated as follows. Let ε be a positive number such that $0 < \varepsilon \leq 1$. By Theorem 3 in (Nelson & Nguyen, 2013), for any subspace $\mathbf{S}$ of dimension d in $\mathbb{R}^n$, $\Pr\big[(1-\varepsilon)\|\mathbf{v}\|_2 \leq \|\mathbf{P}\mathbf{v}\|_2 \leq (1+\varepsilon)\|\mathbf{v}\|_2\big] \geq 1-\delta$ for all $\mathbf{v} \in \mathbf{S}$ if $p \geq \frac{d^2+d}{\delta(2\varepsilon-\varepsilon^2)^2}$.

The algorithmic details of Noisy-DR-ℓ^0-SSC-OSNAP are the same as those of Noisy-DR-ℓ^0-SSC expect for $\mathbf{P}$. Using the ℓ^2-norm preserving property of OSNAP, we have the following theorem about the correctness of Noisy-DR-ℓ^0-SSC-OSNAP where each column of $\mathbf{P}$ only has 1 nonzero element.

Theorem 3.2. (Subspace detection property holds for Noisy-DR-ℓ^0-SSC-OSNAP under deterministic model with $\mathbf{P}$ being the OSNAP) Let nonzero vector $\boldsymbol{\beta}^*$ be the optimal solution to the noisy ℓ^0-SSC problem (3) for point $\mathbf{x}_i$ with $\|\boldsymbol{\beta}^*\|_0 = r^*$, $n_k \geq d_k + 1$ for every $1 \leq k \leq K$, and there exists $1 < r_0 \leq d$ such that $1 < r^* \leq r_0$. Suppose $\mathbf{Y}$ is in general position, $\mathbf{y}_i \in \mathcal{S}_k$ for some $1 \leq k \leq K$, $\delta < \min_{1\leq r\leq r_0} \bar{\sigma}_{\mathbf{Y},r}$. Let $M_{i,\delta} \triangleq M_i - \delta$, $d_{\max} \triangleq \max_k d_k$, ε be a positive number such that $0 < \varepsilon \leq 1$. Suppose

$$M_{i,\delta} > \frac{2(1+\varepsilon)^3\delta}{\sigma_{\mathbf{X},r_0}}, \tag{16}$$

and

$$\mu_{r,\varepsilon} \triangleq \frac{\delta}{\frac{\min_{1\leq r'<r} \bar{\sigma}_{\mathbf{Y},r}}{(1+\varepsilon)^2} - \delta} < 1 - \frac{2(1+\varepsilon)^2\delta}{\sigma_{\mathbf{X},r_0}}. \tag{17}$$

Then if

$$\tilde{\lambda}_0 < \tilde{\lambda} < 1, \tag{18}$$

where $\tilde{\lambda}_0 \triangleq \max\{\lambda_1, \lambda_2, \frac{1}{r_0}\}$ and

$$\lambda_1 \triangleq \inf\{0 < \tilde{\lambda} < 1 \colon \sqrt{1+\varepsilon-\tilde{\lambda}} + \frac{2\delta}{\sigma_{\tilde{\mathbf{X}},r_0}\sqrt{\tilde{\lambda}}} < M_{i,\delta}\}, \tag{19}$$

$$\lambda_2 \triangleq \inf\{0 < \lambda < 1 \colon \lambda - \frac{2\delta}{\sigma_{\tilde{\mathbf{X}},r_0}}\frac{1}{\sqrt{\lambda}} > \tilde{\mu}_{r_0}\}, \tag{20}$$

then with probability at least $1 - K\delta$, the subspace detection property holds for $\tilde{\mathbf{x}}_i$ with $\tilde{\boldsymbol{\beta}}^*$. Here $\tilde{\mu}_{r_0}$ and $\sigma_{\tilde{\mathbf{X}},r_0}$ are defined in (66) and (65) respectively. $\tilde{\boldsymbol{\beta}}^*$ is the optimal solution to (4) with $\mathbf{P}$ being the OSNAP described in the beginning of this subsection with $p \geq \frac{d_{\max}^2+d_{\max}}{\delta(2\varepsilon-\varepsilon^2)^2}$.

3.5 TIME COMPLEXITY OF NOISY ℓ^0-SSC, NOISY-DR-ℓ^0-SSC, NOISY-DR-ℓ^0-SSC-OSNAP

The time complexity of running PGD by Algorithm 1 for noisy ℓ^0-SSC is $\mathcal{O}(Tnd)$, where T is the maximum iteration number. The time complexity of running Algorithm 2 for Noisy-DR-ℓ^0-SSC is comprised of two parts. The first part is the time complexity of steps 1-3 with matrix multiplication and QR decomposition, which is $\mathcal{O}(dp^2 + pdn)$. The second part is the time complexity of step 4, which is $\mathcal{O}(Tnp)$. The overall time complexity of Noisy-DR-ℓ^0-SSC is $\mathcal{O}(dp^2 + pdn + Tnp)$. In practice, p is much smaller than $\min\{d, n, T\}$, so Noisy-DR-ℓ^0-SSC is more efficient than noisy ℓ^0-SSC. Noisy-DR-ℓ^0-SSC-OSNAP is even more efficient than both noisy ℓ^0-SSC and Noisy-DR-ℓ^0-SSC, whose time complexity is $\mathcal{O}(pdn + Tnp)$. This is because the linear transformation $\mathbf{P}$ obtained by OSNAP does require QR decomposition.

3.6 BOUND FOR SUBOPTIMAL AND GLOBALLY OPTIMAL SOLUTIONS FOR NOISY ℓ^0-SSC AND NOISY-DR-ℓ^0-SSC

While our theoretical analysis for noisy ℓ^0-SSC, Noisy-DR-ℓ^0-SSC and Noisy-DR-ℓ^0-SSC is based on optimal solution to the ℓ^0 regularized problem (3), in Section C of the appendix we prove that the bound for the suboptimal solution $\widehat{\boldsymbol{\beta}}$ obtained by Algorithm 1 is in fact close to an optimal solution to (3), justifying the theoretical findings of this paper.

Table 1: Clustering results on various data sets, with the best two results in bold

Data Set	Measure	KM	SC	Noisy SSC	Noisy DR-SSC	SMCE	SSC-OMP	Noisy ℓ^0-SSC	Noisy-DR-ℓ^0-SSC
COIL-20	AC	0.6554	0.4278	0.7854	0.7764	0.7549	0.3389	**0.8472 ± 0.0031**	**0.8479 ± 0.0023**
	NMI	0.7630	0.6217	0.9148	0.9219	0.8754	0.4853	**0.9428 ± 0.0082**	**0.9433 ± 0.0063**
COIL-100	AC	0.4996	0.2835	0.5275	0.5013	0.5639	0.1667	**0.7683** ± 0.0020	**0.7039** ± 0.0087
	NMI	0.7539	0.5923	0.8041	0.8019	0.8064	0.3757	**0.9182** ± 0.0096	**0.8706** ± 0.0109
Yale-B	AC	0.0954	0.1077	0.7850	0.7255	0.3293	0.7789	**0.8480** ± 0.0091	**0.8231** ± 0.0173
	NMI	0.1258	0.1485	0.7760	0.7311	0.3812	0.7024	**0.8612** ± 0.0072	**0.8533** ± 0.0294
MPIE S1	AC	0.1164	0.1285	0.5892	0.3588	0.1721	0.1695	**0.6741**± 0.0413	**0.6741**± 0.0938
	NMI	0.5049	0.5292	0.7653	0.6806	0.5514	0.3395	**0.8622**± 0.0533	**0.8622**± 0.0834
MPIE S2	AC	0.1315	0.1410	0.6994	0.4611	0.1898	0.2093	**0.7527**± 0.0115	**0.7527**± 0.0596
	NMI	0.4834	0.5128	0.8149	0.7086	0.5293	0.4292	**0.8939**± 0.0389	**0.7527** ± 0.0742
MPIE S3	AC	0.1291	0.1459	0.6316	0.4841	0.1856	0.1787	**0.7050**± 0.0277	**0.7050**± 0.0812
	NMI	0.4811	0.5185	0.7858	0.7340	0.5155	0.3415	**0.8750**± 0.0157	**0.8750**± 0.0693
MPIE S4	AC	0.1308	0.1463	0.6803	0.5511	0.1823	0.1680	**0.7246**± 0.0147	**0.7246**± 0.0605
	NMI	0.4866	0.5280	0.8063	0.7955	0.5294	0.3345	**0.8837**± 0.0212	**0.8837**± 0.0781
MNIST	AC	0.5236	0.3504	0.5714	0.5123	**0.6542**	0.5561	0.6259 ± 0.0249	**0.6296** ± 0.152
	NMI	0.4770	0.3607	0.6091	0.5026	**0.6796**	0.5986	**0.6501** ± 0.0196	0.6440 ± 0.0259

4 EXPERIMENTAL RESULTS

4.1 PERFORMANCE OF NOISY-DR-ℓ^0-SSC AND NOISY-DR-ℓ^0-SSC-OSNAP

We first demonstrate the performance of Noisy-DR-ℓ^0-SSC, with comparison to other competing clustering methods including K-Means (KM), Spectral Clustering (SC), noisy SSC, Sparse Manifold Clustering and Embedding (SMCE) (Elhamifar & Vidal, 2011) and SSC-OMP (Dyer et al., 2013) in Table 1. With the coefficient matrix $\mathbf{Z}$ obtained by the optimization of noisy ℓ^0-SSC or Noisy-DR-ℓ^0-SSC, a sparse similarity matrix is built by $\mathbf{W} = \frac{|\mathbf{Z}|+|\mathbf{Z}^\top|}{2}$, and then spectral clustering is performed on $\mathbf{W}$ to obtain the clustering results. Two measures are used to evaluate the performance of different clustering methods, i.e. the Accuracy (AC) and the Normalized Mutual Information (NMI) (Zheng et al., 2004). The introduction to the datasets is in Section D of the appendix.

We use randomized rank-p decomposition of the data matrix in Noisy-DR-ℓ^0-SSC with $p = \frac{\min\{d,n\}}{10}$. It can be observed that noisy ℓ^0-SSC and Noisy-DR-ℓ^0-SSC always achieve better performance than other methods in Table 1, including the noisy SSC on dimensionality reduced data (Noisy DR-SSC) (Wang et al., 2015). Note that noisy ℓ^0-SSC has the same performance as ℓ^0-SSC (Yang et al., 2016). For all the methods that involve random projection, we conduct the experiments for 30 times and report the average performance. Note that the cluster accuracy of SSC-OMP on the extended Yale-B data set (Yale-B) is reported according to (You et al., 2016). We randomly sample 1000 images from each class of the MNIST data set so as to collect a total number of 10000 images on which clustering is performed, and the average performance of 10 random sampling is reported for this data set.

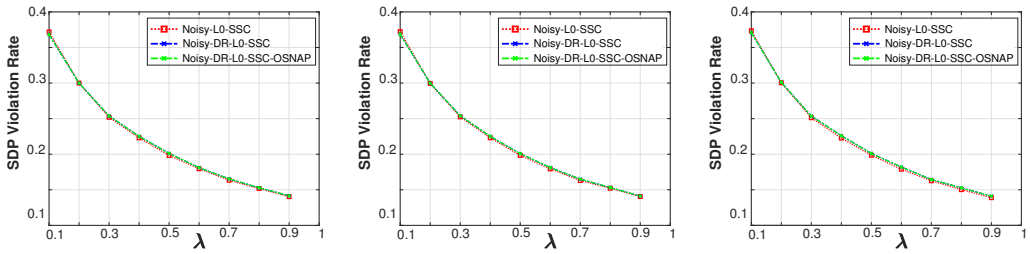
We present more results of Noisy-DR-ℓ^0-SSC and Noisy-DR-ℓ^0-SSC-OSNAP in Table 2 with different values of p. It can be observed that both Noisy-DR-ℓ^0-SSC and Noisy-DR-ℓ^0-SSC-OSNAP always render comparable or even better results on highly compressed data, when compared to noisy ℓ^0-SSC. The actual running time of both algorithms confirms the time complexity analysis in Section 3.5, and we observe that Noisy-DR-ℓ^0-SSC and Noisy-DR-ℓ^0-SSC-OSNAP are 8.7 times and 9.6 times faster than noisy ℓ^0-SSC with the same number of iterations T on the Extended Yale-B (Yale-B) data set. Figure 3 in Section D of the appendix illustrates how the accuracy and NMI vary with respect to λ on Yale-B.

Table 2: Clustering results on various data sets, with different values of p for the linear transformation $\mathbf{P}$ and the best two results in bold

Data Set	Measure	Noisy ℓ^0-SSC	Noisy-DR-ℓ^0-SSC			Noisy-DR-ℓ^0-SSC-OSNAP		
p			$p = \min\{d,n\}/5$	$p = \min\{d,n\}/10$	$p = \min\{d,n\}/15$	$p = \min\{d,n\}/5$	$p = \min\{d,n\}/10$	$p = \min\{d,n\}/15$
COIL-20	AC	0.8472	0.8479	0.8479	**0.8479**	**0.8486**	0.8472	0.8472
	NMI	0.9428	0.9433	0.9433	**0.9433**	**0.9439**	0.9428	0.9428
COIL-100	AC	**0.7683**	0.6992	**0.7276**	0.7043	**0.5404**	0.7046	0.7233
	NMI	**0.9182**	0.8626	**0.8919**	0.8636	0.7819	0.8708	**0.8726**
Yale-B	AC	**0.8480**	0.8219	0.8231	0.8289	**0.8500**	0.8318	0.8277
	NMI	0.8612	0.8519	0.8527	0.8534	0.8538	**0.8593**	**0.8594**

4.2 Justification of Theoretical Analysis for noisy ℓ^0-SSC

We further demonstrates the practical implication of our theoretical Analysis for Noisy-DR-ℓ^0-SSC, Noisy-DR-ℓ^0-SSC-OSNAP, and noisy ℓ^0-SSC. The conditions in terms of λ, (12) of Theorem 3.1 for Noisy-DR-ℓ^0-SSC, (18) of Theorem 3.2 for Noisy-DR-ℓ^0-SSC-OSNAP, and (31) of Theorem A.4 for noisy ℓ^0-SSC together with Remark A.6, unanimously indicate that a relatively large λ preserves the subspace detection property with a high probability. This theoretical finding is consistent with the empirical study in the following experiment. We add Gaussian noise of zero mean and different variance σ^2 to the extended Yale-B data set. Figure 1(a) to Figure 1(c) illustrate SDP violation with respect to λ for different noise levels with σ^2 ranging over $10, 30, 60$. The SDP violation is defined in (Wang & Xu, 2013) which is the percentage of pairs of data points which are mistakenly put in the same subspace by the similarity matrix $\mathbf{W}$, namely the percentage of pairs $(\mathbf{x}_i, \mathbf{x}_j)$ with nonzero $\mathbf{W}_{ij}$ while they are in fact not in the same subspace. It can be observed that increasing λ effectively reduces SDP violation for noisy ℓ^0-SSC, Noisy-DR-ℓ^0-SSC and Noisy-DR-ℓ^0-OSNAP, confirming our theoretical prediction. Section D constains more results about SDP violation rates with respect to λ.



(a) $\sigma^2 = 10$ (b) $\sigma^2 = 30$ (c) $\sigma^2 = 60$

Figure 1: The SDP violation rate with respect to λ for noisy ℓ^0-SSC, Noisy-DR-ℓ^0-SSC and Noisy-DR-ℓ^0-SSC-OSNAP with different noise levels σ^2. The SDP violation rates of Noisy-DR-ℓ^0-SSC and that for Noisy-DR-ℓ^0-SSC-OSNAP are almost the same, so their curves largely overlap each other.

5 Conclusion

In this paper, we first prove that noisy ℓ^0-SSC recovers subspaces from noisy data through ℓ^0-induced sparsity. In order to improve the efficiency of noisy ℓ^0-SSC, we propose Noisy-DR-ℓ^0-SSC and Noisy-DR-ℓ^0-SSC which perform noisy ℓ^0-SSC on dimensionality reduced data and they still provably recover the subspaces in the original data. Experiment results demonstrate the effectiveness of Noisy-DR-ℓ^0-SSC and Noisy-DR-ℓ^0-SSC-OSNAP.

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A THEORETICAL ANALYSIS FOR NOISY ℓ^0-SSC

The theoretical results on the subspace detection property for noisy ℓ^0-SSC are presented in this section under deterministic model and randomized models. These models are introduced in the following subsection.

A.1 MODELS

Similar to (Soltanolkotabi & Candés, 2012), we introduce the deterministic, semi-random and fully-random models for the analysis of noisy ℓ^0-SSC.

- **Deterministic Model:** the subspaces and the data in each subspace are fixed.
- **Semi-Random Model:** the subspaces are fixed but the data are independent and identically distributed in each of the subspaces.
- **Fully-Random Model:** both the subspaces and the data of each subspace are independent and identically distributed.

The data in the above definitions refer to clean data without noise. We refer to semi-random model and fully-random model as randomized models in this paper. All the three models are extensively employed to analyze the subspace detection property in the subspace learning literature (Soltanolkotabi & Candés, 2012; Wang et al., 2013; Wang & Xu, 2013; Yining Wang & Singh, 2016).

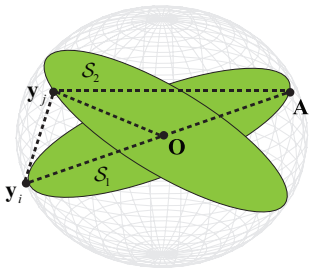


Figure 2: Illustration of an external subspace. All the data $\mathbf{Y}$ are normalized to have unit norm for illustration purpose, so they lie on the surface of the sphere. $\mathcal{S}_1$ and $\mathcal{S}_2$ are two subspaces in the three-dimensional ambient space. The subspace spanned by $\mathbf{y}_i \in \mathcal{S}_1$ and $\mathbf{y}_j \in \mathcal{S}_2$ is an external subspace, and the intersection of this external subspace and $\mathcal{S}_1$ is a dashed line $\mathbf{y}_i \mathbf{O} A$.

A.2 NOISY ℓ^0-SSC: DETERMINISTIC ANALYSIS

We introduce the definition of general position and external subspace before our analysis on noisy ℓ^0-SSC.

Definition A.1. (General position) For any $1 \leq k \leq K$, the data $\mathbf{Y}^{(k)}$ are in general position if any subset of $L \leq d_k$ data points (columns) of $\mathbf{Y}^{(k)}$ are linearly independent. $\mathbf{Y}$ are in general position if $\mathbf{Y}^{(k)}$ are in general position for $1 \leq k \leq K$.

The assumption of general condition is rather mild. In fact, if the data points in $\mathbf{X}^{(k)}$ are independently distributed according to any continuous distribution, then they almost surely in general position.

Let the distance between a point $\mathbf{x} \in \mathbb{R}^d$ and a subspace $\mathcal{S} \subseteq \mathbb{R}^d$ be defined as $d(\mathbf{x}, \mathcal{S}) = \inf_{\mathbf{y} \in \mathcal{S}} \|\mathbf{x} - \mathbf{y}\|_2$, the definition of external subspaces is presented as follows. Figure 2 illustrates an example of external subspace.

Definition A.2. (External subspace) For a point $\mathbf{y} \in \mathbf{Y}^{(k)}$, a subspace $\mathbf{H}_{\{\mathbf{y}_{i_j}\}_{j=1}^L}$ spanned by a set of linear independent points $\{\mathbf{y}_{i_j}\}_{j=1}^L \subseteq \mathbf{Y}$ is defined to be an external subspace of $\mathbf{y}$ if

$\{\mathbf{y}_{i_j}\}_{j=1}^{L} \not\subseteq \mathbf{Y}^{(k)}$ and $\mathbf{y} \notin \{\mathbf{y}_{i_j}\}_{j=1}^{L}$. The point $\mathbf{y}$ is said to be away from its external subspaces if $\min_{\mathbf{H}\in\mathcal{H}_{\mathbf{y},d_k}} d(\mathbf{y},\mathbf{H}) > 0$, where $\mathcal{H}_{\mathbf{y},d}$ are the set of all external subspaces of $\mathbf{y}$ of dimension no greater than d for $\mathbf{y}$, i.e. $\mathcal{H}_{\mathbf{y},d} = \{\mathbf{H}\colon \mathbf{H} = \mathbf{H}_{\{\mathbf{y}_{i_j}\}_{j=1}^{L}}, \dim[\mathbf{H}] = L, L \le d, \{\mathbf{y}_{i_j}\}_{j=1}^{L} \not\subseteq \mathbf{Y}^{(k)}, \mathbf{y} \notin \{\mathbf{y}_{i_j}\}_{j=1}^{L}\}$. All the data points in $\mathbf{Y}^{(k)}$ are said to be away from the external subspaces if each of them is away from the its associated external spaces.

Remark A.1. (Subspace detection property holds for noiseless ℓ^0-SSC under the deterministic model) It can be verified that the following statement is true. Under the deterministic model, suppose data is noiseless, $n_k \ge d_k + 1$, $\mathbf{Y}^{(k)}$ is in general position. If all the data points in $\mathbf{Y}^{(k)}$ are away from the external subspaces for any $1 \le k \le K$, then the subspace detection property for ℓ^0-SSC holds with an optimal solution $\mathbf{Z}^*$ to (1).

To present our theoretical results of the correctness of noisy ℓ^0-SSC, we need the definitions related to the spectrum of $\mathbf{X}$ and $\mathbf{Y}$, which are defined as follows. In the following analysis, we employ $\boldsymbol{\beta}$ to denote the sparse code of datum $\mathbf{x}_i$ so that a simpler notation other than $\mathbf{Z}^i$ is dedicated to our analysis.

Definition A.3. The minimum restricted eigenvalue of the uncorrupted data is defined as

$$\sigma_{\mathbf{Y},r} \triangleq \min_{\boldsymbol{\beta}:\|\boldsymbol{\beta}\|_0=r,\mathrm{rank}(\mathbf{Y}_{\boldsymbol{\beta}})=\|\boldsymbol{\beta}\|_0} \sigma_{\min}(\mathbf{Y}_{\boldsymbol{\beta}}) \tag{21}$$

for $r \ge 1$. In addition, the normalized minimum restricted eigenvalue of the uncorrupted data is defined by

$$\bar{\sigma}_{\mathbf{Y},r} \triangleq \frac{\sigma_{\mathbf{Y},r}}{\sqrt{r}} \tag{22}$$

Moreover, the following quantities are defined for our analysis. We define

$$\tau_0 \triangleq \frac{2\delta\sqrt{r^*}}{\sigma^*_{\mathbf{X}}} + \tau_1, \tag{23}$$

where

$$\tau_1 \triangleq \frac{\delta}{\bar{\sigma}^*_{\mathbf{Y}} - \delta}, \quad \sigma^*_{\mathbf{X}} \triangleq \sigma_{\min}(\mathbf{X}_{\boldsymbol{\beta}^*}), \tag{24}$$

with $\delta < \bar{\sigma}^*_{\mathbf{Y}}$, and $\bar{\sigma}^*_{\mathbf{Y}}$ is defined as

$$\bar{\sigma}^*_{\mathbf{Y}} \triangleq \min_{1\le r<r^*} \bar{\sigma}_{\mathbf{Y},r}, \tag{25}$$

Now we present the main result on the correctness of noisy ℓ^0-SSC.

Theorem A.2. (Subspace detection property holds for noisy ℓ^0-SSC) Let nonzero vector $\boldsymbol{\beta}^*$ be an optimal solution to the noisy ℓ^0-SSC problem (3) for point $\mathbf{x}_i$ with $\|\boldsymbol{\beta}^*\|_0 = r^* > 1$, and $c^* \triangleq \|\mathbf{x}_i - \mathbf{X}\boldsymbol{\beta}^*\|_2$. Suppose $\mathbf{Y}$ is in general position, $\mathbf{y}_i \in \mathcal{S}_k$ for some $1 \le k \le K$, $\delta < \bar{\sigma}^*_{\mathbf{Y}}$, $\lambda > \tau_0$, $\mathbf{B}(\mathbf{y}_i, \delta + c^* + \frac{2\delta\sqrt{r^*}}{\sigma^*_{\mathbf{X}}}) \cap \mathbf{H} = \emptyset$ for any $\mathbf{H} \in \mathcal{H}_{\mathbf{y}_i,d_k}$. Then the subspace detection property holds for $\mathbf{x}_i$ with $\boldsymbol{\beta}^*$. Here τ_0, τ_1, $\bar{\sigma}^*_{\mathbf{Y}}$ and $\sigma^*_{\mathbf{X}}$ are defined in (23), (24) and (25).

Remark A.3. When $\delta = 0$ and there is no noise in the data $\mathbf{X}$, the conditions for the correctness of noisy ℓ^0-SSC in Theorem A.2 almost reduce to that for noiseless ℓ^0-SSC. To see this, the conditions are reduced to $\mathbf{B}(\mathbf{y}_i, c^*) \cap \mathbf{H} = \emptyset$, which are exactly the conditions required by noiseless ℓ^0-SSC, namely data are away from the external subspaces by choosing $\lambda \to 0$ and it follows that $c^* = 0$.

While Theorem A.2 establishes geometric conditions under which the subspace detection property holds for noisy ℓ^0-SSC, it can be seen that these conditions are often coupled with an optimal solution $\boldsymbol{\beta}^*$ to the noisy ℓ^0-SSC problem (3). In the following theorem, the correctness of noisy ℓ^0-SSC is guaranteed in terms of λ, the weight for the ℓ^0 regularization term in (3), and the geometric conditions independent of an optimal solution to (3).

Let $M_i > 0$ be the minimum distance between $\mathbf{y}_i \in \mathcal{S}_k$ and its external subspaces when $\mathbf{y}_i$ is away from its external subspaces, i.e.

$$M_i \triangleq \min\{d(\mathbf{y}_i, \mathbf{H})\colon \mathbf{H} \in \mathcal{H}_{\mathbf{y}_i,d_k}\}, \tag{26}$$

The following two quantities related to the spectrum of clean and noisy data, μ_r and $\sigma_{\mathbf{X},r}$, are defined as follows with $r > 1$ for the analysis in Theorem A.4.

$$\mu_r \triangleq \frac{\delta}{\min_{1\leq r'<r} \bar{\sigma}_{\mathbf{Y},r} - \delta}, \tag{27}$$

$$\sigma_{\mathbf{X},r} \triangleq \min\{\sigma_{\min}(\mathbf{X}_{\boldsymbol{\beta}})\colon 1 \leq \|\boldsymbol{\beta}\|_0 \leq r\} \tag{28}$$

Theorem A.4. (Subspace detection property holds for noisy ℓ^0-SSC under deterministic model, with conditions in terms of λ) Let nonzero vector $\boldsymbol{\beta}^*$ be an optimal solution to the noisy ℓ^0-SSC problem (3) for point $\mathbf{x}_i$ with $\|\boldsymbol{\beta}^*\|_0 = r^*$, $n_k \geq d_k + 1$ for every $1 \leq k \leq K$, and there exists $1 < r_0 \leq d$ such that $1 < r^* \leq r_0$. Suppose $\mathbf{Y}$ is in general position, $\mathbf{y}_i \in \mathcal{S}_k$ for some $1 \leq k \leq K$, $\delta < \min_{1\leq r<r_0} \bar{\sigma}_{\mathbf{Y},r}$, and $M_{i,\delta} \triangleq M_i - \delta$. Suppose

$$M_{i,\delta} > \frac{2\delta}{\sigma_{\mathbf{X},r_0}}, \tag{29}$$

and

$$\mu_{r_0} < 1 - \frac{2\delta}{\sigma_{\mathbf{X},r_0}}. \tag{30}$$

Then if

$$\lambda_0 < \lambda < 1, \tag{31}$$

where $\lambda_0 \triangleq \max\{\lambda_1, \lambda_2\}$ and

$$\lambda_1 \triangleq \inf\{0 < \lambda < 1\colon \sqrt{1-\lambda} + \frac{2\delta}{\sigma_{\mathbf{X},r_0}\sqrt{\lambda}} < M_{i,\delta}\}, \tag{32}$$

$$\lambda_2 \triangleq \inf\{0 < \lambda < 1\colon \lambda - \frac{2\delta}{\sigma_{\mathbf{X},r_0}}\frac{1}{\sqrt{\lambda}} > \mu_{r_0}\}, \tag{33}$$

the subspace detection property holds for $\mathbf{x}_i$ with $\boldsymbol{\beta}^*$. Here M_i, μ_{r_0} and $\sigma_{\mathbf{X},r_0}$ are defined in (26), (27) and (28) respectively.

Remark A.5. The two conditions (29) and (30) are induced by two conditions, $\mathbf{B}(\mathbf{y}_i, \delta + c^* + \frac{2\delta\sqrt{r^*}}{\sigma^*_{\mathbf{X}}}) \cap \mathbf{H} = \emptyset$ for any $\mathbf{H} \in \mathcal{H}_{\mathbf{y}_i,d_k}$ and $\lambda > \tau_0$ respectively, which are required by Theorem A.2. Note that when (29) and (30) hold, λ_1 and λ_2 can always be chosen in accordance with (32) and (33).

Remark A.6. It can be observed from condition (31) that noisy ℓ^0-SSC encourages sparse solution by a relatively large λ so as to guarantee the subspace detection property. This theoretical finding is consistent with the empirical study shown in the experimental results.

A.3 Noisy ℓ^0-SSC: Randomized Analysis

In this subsection, the correctness of noisy ℓ^0-SSC is analyzed when the clean data in each subspace are distributed at random. We assume that the data in subspace $\mathcal{S}^{(k)}$ are i.i.d. according to uniform distribution on the unit sphere centered at the origin, for $1 \leq k \leq K$. This setting is employed extesively in the subspace learning literature (Soltanolkotabi & Candés, 2012; Wang et al., 2013; Wang & Xu, 2013; Yining Wang & Singh, 2016).

We then have the following results regarding to the subspace detection property of noisy ℓ^0-SSC under randomized models.

Theorem A.7. (Subspace detection property holds for noisy ℓ^0-SSC under randomized models, with conditions in terms of λ) Under randomized models, let nonzero vector $\boldsymbol{\beta}^*$ be an optimal solution to the noisy ℓ^0-SSC problem (3) for point $\mathbf{x}_i$ with $\|\boldsymbol{\beta}^*\|_0 = r^*$, $n_k \geq d_k + 1$ for every $1 \leq k \leq K$, and there exists $1 < r_0 \leq d$ such that $1 < r^* \leq r_0$. Suppose the data in each subspace are i.i.d. isotropic samples according to some continuous distribution that satisfies condition (a). Let $d_{\max} \triangleq \max_k d_k$, $\sigma'_{\min} \triangleq \frac{1}{\sqrt{d_{\max}}}\big(1 - 14\sqrt[3]{4}(\frac{M^2 d}{n})^{2/3}\big)$, $c \triangleq \sqrt{\frac{\sigma'^2_{\min} - r_0 \operatorname{aff}(\mathcal{S}_{t_1}, \mathcal{S}_{t_2})}{r_0}}$. For $t > 0$ such that $\frac{1}{d_{\max}} - 2t\sqrt{d_{\max} - 1} - t^2 > 0$, suppose

$$\max_{t_1 \neq t_2} \operatorname{aff}(\mathcal{S}_{t_1}, \mathcal{S}_{t_2}) < \frac{\sigma'^2_{\min}}{r_0}, \tag{34}$$

$$\delta < c, \tag{35}$$

$$\delta + \frac{2\delta}{\sqrt{r_0}(c-\delta)} \leq \frac{1}{d_{\max}} - 2t\sqrt{d_{\max}-1} - t^2, \tag{36}$$

$$\frac{\delta}{c-\delta} + \frac{2\delta}{\sqrt{r_0}(c-\delta)} < 1, \tag{37}$$

and

$$\lambda_0' < \lambda < 1, \tag{38}$$

where $M \triangleq \sup_{0\leq t<1} \frac{2t^3 \arccos t}{\pi} < 1$, $\lambda_0' \triangleq \max\{\lambda_1', \lambda_2'\}$ and

$$\lambda_1' \triangleq \inf\{0 < \lambda < 1 \colon \sqrt{1-\lambda} + \frac{2\delta}{\sqrt{r_0}(c-\delta)\sqrt{\lambda}} < \frac{1}{d_{\max}} - 2t\sqrt{d_{\max}-1} - t^2 - \delta\}, \tag{39}$$

$$\lambda_2' \triangleq \inf\{0 < \lambda < 1 \colon \lambda - \frac{2\delta}{\sqrt{r_0}(c-\delta)}\frac{1}{\sqrt{\lambda}} > \frac{\delta}{c-\delta}\}. \tag{40}$$

Then with probability at least $1 - K\exp(-d) - 8\sum_{k=1}^{K} n_k \exp(-\frac{d_k t^2}{2})$, the subspace detection property holds for $\mathbf{x}_i$ with $\boldsymbol{\beta}^*$.

Remark A.8. Note that there is no assumption on the distribution of subspaces in Theorem A.7, so it is not required that the subspaces should have uniform distribution, an is required in the geometric analysis of ℓ^1-SSC (Soltanolkotabi & Candés, 2012) and its noisy version (Wang & Xu, 2013).

B PROOFS

We provide proofs to the lemmas and theorems in the paper in this subsection.

B.1 PROOF OF REMARK A.1

Lemma A. (Subspace detection property holds for ℓ^0-SSC under the deterministic model) Under the deterministic model, suppose data is noiseless, $n_k \geq d_k + 1$, $\mathbf{Y}^{(k)}$ is in general position. If all the data points in $\mathbf{Y}^{(k)}$ are away from the external subspaces for any $1 \leq k \leq K$, then the subspace detection property for ℓ^0-SSC holds with an optimal solution $\mathbf{Z}^*$ to (1).

Proof. Let $\mathbf{x}_i \in \mathcal{S}_k$. Note that $\mathbf{Z}^{*i}$ is an optimal solution to the following ℓ^0 sparse representation problem

$$\min_{\mathbf{Z}^i} \|\mathbf{Z}^i\|_0 \quad s.t.\ \mathbf{x}_i = [\mathbf{X}^{(k)} \setminus \mathbf{x}_i \quad \mathbf{X}^{(-k)}]\mathbf{Z}^i,\ \mathbf{Z}_{ii} = 0, \tag{41}$$

where $\mathbf{X}^{(-k)}$ denotes the data that lie in all subspaces except $\mathcal{S}_k$. Let $\mathbf{Z}^{*i} = \begin{bmatrix} \boldsymbol{\alpha} \\ \boldsymbol{\beta} \end{bmatrix}$ where $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ are sparse codes corresponding to $\mathbf{X}^{(k)} \setminus \mathbf{x}_i$ and $\mathbf{X}^{(-k)}$ respectively.

Suppose $\boldsymbol{\beta} \neq \mathbf{0}$, then $\mathbf{x}_i$ belongs to a subspace $\mathcal{S}^{'} = \mathbf{H}_{\mathbf{X}_{\mathbf{Z}^{*i}}}$ spanned by the projected data points corresponding to nonzero elements of $\mathbf{Z}^{*i}$, and $\mathcal{S}^{'} \neq \mathcal{S}_k$, $\dim[\mathcal{S}^{'}] \leq d_k$. To see this, if $\mathcal{S}^{'} = \mathcal{S}_k$, then the data corresponding to nonzero elements of $\boldsymbol{\beta}$ belong to $\mathcal{S}_k$, which is contrary to the definition of $\mathbf{X}^{(-k)}$. Also, if $\dim[\mathcal{S}^{'}] > d_k$, then any d_k points in $\mathbf{X}^{(k)}$ can be used to linearly represent $\mathbf{x}_i$ by the condition of general position, contradicting with the optimality of $\mathbf{Z}^{*i}$. Since the data points (or columns) in $\mathbf{X}_{\mathbf{Z}^{*i}}$ are linearly independent, it follows that $\mathbf{x}_i$ lies in an external subspace $\mathbf{H}_{\mathbf{X}_{\mathbf{Z}^{*i}}}$ spanned by linearly independent points in $\mathbf{X}_{\mathbf{Z}^{*i}}$, and $\dim[\mathbf{H}_{\mathbf{X}_{\mathbf{Z}^{*i}}}] = \dim[\mathcal{S}^{'}] \leq d_k$. This contradicts with the assumption that $\mathbf{x}_i$ is away from the external subspaces. Therefore, $\boldsymbol{\beta} = \mathbf{0}$. Perform the above analysis for all $1 \leq i \leq n$, we can prove that the subspace detection property holds for all $1 \leq i \leq n$.

□

B.2 PROOF OF THEOREM A.2

Before proving this theorem, we introduce the following perturbation bound for the distance between a data point and the subspaces spanned by noisy and noiseless data, which is useful to establish the conditions when the subspace detection property holds for noisy ℓ^0-SSC.

Lemma B.1. Let $\boldsymbol{\beta} \in \mathbb{R}^n$ and $\mathbf{Y}_{\boldsymbol{\beta}}$ has full column rank. Suppose $\delta < \bar{\sigma}_{\mathbf{Y},r}$ where $r = \|\boldsymbol{\beta}\|_0$, then $\mathbf{X}_{\boldsymbol{\beta}}$ is a full column rank matrix, and

$$|d(\mathbf{x}_i, \mathbf{H}_{\mathbf{X}_{\boldsymbol{\beta}}}) - d(\mathbf{x}_i, \mathbf{H}_{\mathbf{Y}_{\boldsymbol{\beta}}})| \leq \frac{\delta}{\bar{\sigma}_{\mathbf{Y},r} - \delta} \tag{42}$$

for any $1 \leq i \leq n$.

Lemma B.2 shows that an optimal solution to the noisy ℓ^0-SSC problem (3) is also that to a ℓ^0-minimization problem with tolerance to noise.

Lemma B.2. Let nonzero vector $\boldsymbol{\beta}^*$ be an optimal solution to the noisy ℓ^0-SSC problem (3) for point $\mathbf{x}_i$ with $\|\boldsymbol{\beta}^*\|_0 = r^* > 1$. If $\lambda > \tau_0$ where τ_0 is defined as

$$\tau_0 \triangleq \frac{2\delta\sqrt{r^*}}{\sigma^*_{\mathbf{X}}} + \tau_1,$$

where

$$\tau_1 \triangleq \frac{\delta}{\bar{\sigma}^*_{\mathbf{Y}} - \delta}, \quad \sigma^*_{\mathbf{X}} \triangleq \sigma_{\min}(\mathbf{X}_{\boldsymbol{\beta}^*}),$$

with $\delta < \bar{\sigma}^*_{\mathbf{Y}}$, and $\bar{\sigma}^*_{\mathbf{Y}}$ is defined as

$$\bar{\sigma}^*_{\mathbf{Y}} \triangleq \min_{1 \leq r < r^*} \bar{\sigma}_{\mathbf{Y},r},$$

then $\boldsymbol{\beta}^*$ is an optimal solution to the following sparse approximation problem with the uncorrupted data as the dictionary:

$$\min_{\boldsymbol{\beta}} \|\boldsymbol{\beta}\|_0 \quad s.t.\ \|\mathbf{x}_i - \mathbf{Y}\boldsymbol{\beta}\|_2 \leq c^* + \frac{2\delta\sqrt{r^*}}{\sigma^*_{\mathbf{X}}},\ \boldsymbol{\beta}_i = 0. \tag{43}$$

where $c^* \triangleq \|\mathbf{x}_i - \mathbf{X}\boldsymbol{\beta}^*\|_2$.

Define $\mathbf{B}(\mathbf{x}_i, c_0) = \{\mathbf{x} \colon \|\mathbf{x} - \mathbf{x}_i\| \leq c_0\}$ be the ball centered at $\mathbf{x}_i$ with radius c_0. If $\mathbf{B}(\mathbf{x}_i, c_0)$ is away from the corresponding confusion area, i.e. all the external subspaces in $\mathcal{H}_{\mathbf{y}_i,d_k}$, then subspace detection property holds with the solution to a proper sparse approximation problem where $\mathbf{x}_i$ is approximated by the uncorrupted data, as shown in the following Lemma.

Lemma B.3. Suppose $\mathbf{Y}$ is in general position and $\mathbf{y}_i \in \mathcal{S}_k$ for some $1 \leq k \leq K$. For positive number c_0 such that $c_0 \geq d(\mathbf{x}_i, \mathcal{S}_k)$, suppose $\mathbf{B}(\mathbf{x}_i, c_0) \cap \mathbf{H} = \emptyset$ for any $\mathbf{H} \in \mathcal{H}_{\mathbf{y}_i,d_k}$. Then the subspace detection property holds for $\mathbf{x}_i$ with an optimal solution to the following sparse approximation problem, denoted by $\boldsymbol{\beta}^*$, i.e. nonzero elements of $\boldsymbol{\beta}^*$ correspond to the columns of $\mathbf{X}$ from the same subspace as $\mathbf{y}_i$.

$$\min_{\boldsymbol{\beta}} \|\boldsymbol{\beta}\|_0 \quad s.t.\ \|\mathbf{x}_i - \mathbf{Y}\boldsymbol{\beta}\|_2 \leq c_0,\ \boldsymbol{\beta}_i = 0. \tag{44}$$

Now we are ready to prove Theorem A.2.

Proof of Theorem A.2. We first show that $d(\mathbf{x}_i, \mathcal{S}_k) \leq c^* + \frac{2\delta\sqrt{r^*}}{\sigma^*_{\mathbf{X}}}$. To see this, $\sigma^*_{\mathbf{X}} = \sigma_{\min}(\mathbf{X}_{\boldsymbol{\beta}^*}) \leq 1$ as the columns of $\mathbf{X}$ have unit ℓ^2-norm. It follows that

$$c^* + \frac{2\delta\sqrt{r^*}}{\sigma^*_{\mathbf{X}}} \geq 2\delta\sqrt{r^*} \leq 2\delta > \|\mathbf{x}_i - \mathbf{y}_i\| \leq d(\mathbf{x}_i, \mathcal{S}_k) \tag{45}$$

By Lemma B.2, it can be verified that $\boldsymbol{\beta}^*$ is an optimal solution to the following problem

$$\min_{\boldsymbol{\beta}} \|\boldsymbol{\beta}\|_0 \quad s.t.\ \|\mathbf{x}_i - \mathbf{Y}\boldsymbol{\beta}\|_2 \leq c^* + \frac{2\delta\sqrt{r^*}}{\sigma^*_{\mathbf{X}}},\ \boldsymbol{\beta}_i = 0. \tag{46}$$

The subspace detection property holds which follows from applying Lemma B.3 with $c_0 = c^* + \frac{2\delta\sqrt{r^*}}{\sigma^*_{\mathbf{X}}}$. □

B.3 PROOF OF LEMMA B.1

The following proposition is used for proving Lemma B.1.

Lemma B.4. (Perturbation of distance to subspaces) Let $\mathbf{A}$, $\mathbf{B} \in \mathbb{R}^{m\times n}$ are two matrices and $\text{rank}(\mathbf{A}) = r$, $\text{rank}(\mathbf{B}) = s$. Also, $\mathbf{E} = \mathbf{A} - \mathbf{B}$ and $\|\mathbf{E}\|_2 \leq C$, where $\|\cdot\|_2$ indicates the spectral norm. Then for any point $\mathbf{x} \in \mathbb{R}^m$, the difference of the distance of $\mathbf{x}$ to the column space of $\mathbf{A}$ and $\mathbf{B}$, i.e. $|d(\mathbf{x}, \mathbf{H_A}) - d(\mathbf{x}, \mathbf{H_B})|$, is bounded by

$$|d(\mathbf{x}, \mathbf{H_A}) - d(\mathbf{x}, \mathbf{H_B})| \leq \frac{C\|\mathbf{x}\|_2}{\min\{\sigma_r(\mathbf{A}), \sigma_s(\mathbf{B})\}}. \tag{47}$$

Proof. Note that the projection of $\mathbf{x}$ onto the subspace $\mathbf{H_A}$ is $\mathbf{AA}^+\mathbf{x}$ where $\mathbf{A}^+$ is the Moore-Penrose pseudo-inverse of the matrix $\mathbf{A}$, so $d(\mathbf{x}, \mathbf{H_A})$ equals to the distance between $\mathbf{x}$ and its projection, namely $d(\mathbf{x}, \mathbf{H_A}) = \|\mathbf{x} - \mathbf{AA}^+\mathbf{x}\|_2$. Similarly, $d(\mathbf{x}, \mathbf{H_B}) = \|\mathbf{x} - \mathbf{BB}^+\mathbf{x}\|_2$.

It follows that

$$\begin{aligned} &|d(\mathbf{x}, \mathbf{H_A}) - d(\mathbf{x}, \mathbf{H_B})| = |\|\mathbf{x} - \mathbf{AA}^+\mathbf{x}\|_2 - \|\mathbf{x} - \mathbf{BB}^+\mathbf{x}\|_2| \\ &\leq \|\mathbf{AA}^+\mathbf{x} - \mathbf{BB}^+\mathbf{x}\|_2 \leq \|\mathbf{AA}^+ - \mathbf{BB}^+\|_2\|\mathbf{x}\|_2. \end{aligned} \tag{48}$$

According to the perturbation bound on the orthogonal projection in (Chen et al., 2016; Stewart, 1977),

$$\|\mathbf{AA}^+ - \mathbf{BB}^+\|_2 \leq \max\{\|\mathbf{EA}^+\|_2, \|\mathbf{EB}^+\|_2\}. \tag{49}$$

Since $\|\mathbf{EA}^+\|_2 \leq \|\mathbf{E}\|_2\|\mathbf{A}^+\|_2 \leq \frac{C}{\sigma_r(\mathbf{A})}$, $\|\mathbf{EB}^+\|_2 \leq \|\mathbf{E}\|_2\|\mathbf{B}^+\|_2 \leq \frac{C}{\sigma_s(\mathbf{B})}$, combining (48) and (49), we have

$$\begin{aligned} |d(\mathbf{x}, \mathbf{H_A}) - d(\mathbf{x}, \mathbf{H_B})| &\leq \max\{\frac{C}{\sigma_r(\mathbf{A})}, \frac{C}{\sigma_s(\mathbf{B})}\}\|\mathbf{x}\|_2 \\ &= \frac{C\|\mathbf{x}\|_2}{\min\{\sigma_r(\mathbf{A}), \sigma_s(\mathbf{B})\}}. \end{aligned} \tag{50}$$

So that (42) is proved. □

Proof of Lemma B.1. We have $\mathbf{y}_i = \mathbf{x}_i - \mathbf{n}_i$, and $\sigma_{\min}(\mathbf{Y}_{\boldsymbol{\beta}}^\top \mathbf{Y}_{\boldsymbol{\beta}}) = \left(\sigma_{\min}(\mathbf{Y}_{\boldsymbol{\beta}})\right)^2 \geq \sigma_{\mathbf{Y},r}^2$.

By Weyl (Weyl, 1912), $|\sigma_i(\mathbf{Y}_{\boldsymbol{\beta}}) - \sigma_i(\mathbf{X}_{\boldsymbol{\beta}})| \leq \|\mathbf{N}_{\boldsymbol{\beta}}\|_2 \leq \|\mathbf{N}_{\boldsymbol{\beta}}\|_F \leq \sqrt{r}\delta$. Since $\sqrt{r}\delta < \sigma_{\mathbf{Y},r} \leq \sigma_{\min}(\mathbf{Y}_{\boldsymbol{\beta}}) \leq \sigma_i(\mathbf{Y}_{\boldsymbol{\beta}})$, $\sigma_i(\mathbf{X}_{\boldsymbol{\beta}}) \geq \sigma_i(\mathbf{Y}_{\boldsymbol{\beta}}) - \sqrt{r}\delta \geq \sigma_{\mathbf{Y},r} - \sqrt{r}\delta > 0$ for $1 \leq i \leq \min\{d, r\}$. It follows that $\sigma_{\min}(\mathbf{X}_{\boldsymbol{\beta}}) \geq \sigma_{\mathbf{Y},r} - \sqrt{r}\delta > 0$ and $\mathbf{X}_{\boldsymbol{\beta}}$ has full column rank.

Also, $\|\mathbf{X}_{\boldsymbol{\beta}} - \mathbf{Y}_{\boldsymbol{\beta}}\|_2 \leq \|\mathbf{X}_{\boldsymbol{\beta}} - \mathbf{Y}_{\boldsymbol{\beta}}\|_F \leq \sqrt{r}\delta$. According to Lemma B.4,

$$\begin{aligned} &|d(\mathbf{x}_i, \mathbf{H}_{\mathbf{X}_{\boldsymbol{\beta}}}) - d(\mathbf{x}_i, \mathbf{H}_{\mathbf{Y}_{\boldsymbol{\beta}}})| \\ &\leq \frac{\sqrt{r}\delta}{\min\{\sigma_{\min}(\mathbf{X}_{\boldsymbol{\beta}}), \sigma_{\min}(\mathbf{Y}_{\boldsymbol{\beta}})\}} \\ &\leq \frac{\sqrt{r}\delta}{\sigma_{\mathbf{Y},r} - \sqrt{r}\delta} = \frac{\delta}{\bar{\sigma}_{\mathbf{Y},r} - \delta}. \end{aligned} \tag{51}$$

□

B.4 PROOF OF LEMMA B.2

Proof of Lemma B.2. We have

$$\begin{aligned} &\|\mathbf{x}_i - \mathbf{X}\boldsymbol{\beta}^*\|_2^2 + \lambda\|\boldsymbol{\beta}^*\|_0 \leq \|\mathbf{x}_i - \mathbf{X}\mathbf{0}\|_2^2 + \lambda\|\mathbf{0}\|_0 = 1 \\ &\Rightarrow c^* = \|\mathbf{x}_i - \mathbf{X}\boldsymbol{\beta}^*\|_2 < 1. \end{aligned}$$

We first prove that $\boldsymbol{\beta}^*$ is an optimal solution to the sparse approximation problem

$$\min_{\boldsymbol{\beta}} \|\boldsymbol{\beta}\|_0 \quad s.t.\ \|\mathbf{x}_i - \mathbf{X}\boldsymbol{\beta}\|_2 \leq c^*,\ \boldsymbol{\beta}_i = 0. \tag{52}$$

To see this, suppose there is a vector $\boldsymbol{\beta}'$ such that $\|\mathbf{x}_i - \mathbf{X}\boldsymbol{\beta}'\|_2 \leq c^*$ and $\|\boldsymbol{\beta}'\|_0 < \|\boldsymbol{\beta}^*\|_0$, then $L(\boldsymbol{\beta}') < c^* + \lambda\|\boldsymbol{\beta}^*\|_0 = L(\boldsymbol{\beta}^*)$, contradicting the fact that $\boldsymbol{\beta}^*$ is an optimal solution to (3).

Note that $\mathbf{X}_{\boldsymbol{\beta}^*}$ is a full column rank matrix, otherwise a sparser solution to (3) can be obtained as vector whose support corresponds to the maximal linear independent set of columns of $\mathbf{X}_{\boldsymbol{\beta}^*}$.

Also, the distance between $\mathbf{x}_i$ and the subspace spanned by columns of $\mathbf{X}_{\boldsymbol{\beta}^*}$ equals to c^*, i.e. $d(\mathbf{x}_i, \mathbf{H}_{\mathbf{X}_{\boldsymbol{\beta}^*}}) = c^*$. To see this, it is clear that $d(\mathbf{x}_i, \mathbf{H}_{\mathbf{X}_{\boldsymbol{\beta}^*}}) \leq c^*$. If there is a vector $\mathbf{y} = \mathbf{X}\tilde{\boldsymbol{\beta}}$ in $\mathbf{H}_{\mathbf{X}_{\boldsymbol{\beta}^*}}$ with $\mathrm{supp}(\tilde{\boldsymbol{\beta}}) \subseteq \mathrm{supp}(\boldsymbol{\beta}^*)$, and $\|\mathbf{x}_i - \mathbf{y}\|_2 < c^*$, then $L(\tilde{\boldsymbol{\beta}}) < L(\boldsymbol{\beta}^*)$ which contradicts the optimality of $\boldsymbol{\beta}^*$. Therefore, $d(\mathbf{x}_i, \mathbf{H}_{\mathbf{X}_{\boldsymbol{\beta}^*}}) \geq c^*$, and it follows that $d(\mathbf{x}_i, \mathbf{H}_{\mathbf{X}_{\boldsymbol{\beta}^*}}) = c^*$.

To prove that the subspace separation margin $\mathrm{HS}(\mathbf{x}_i, \mathbf{X}, \boldsymbol{\beta}^*) > 0$, suppose $\mathrm{HS}(\mathbf{x}_i, \mathbf{X}, \boldsymbol{\beta}^*) \leq 0$, so there exists $\boldsymbol{\beta}'$ such that $\|\boldsymbol{\beta}'\|_0 < r^*$, $\mathrm{rank}(\mathbf{X}_{\boldsymbol{\beta}'}) = \|\boldsymbol{\beta}'\|_0$ and $d(\mathbf{y}_i, \mathbf{H}_{\mathbf{X}_{\boldsymbol{\beta}'}}) \leq d(\mathbf{y}_i, \mathbf{H}_{\mathbf{X}_{\boldsymbol{\beta}^*}}) \leq c^*$. Then $\boldsymbol{\beta}'$ is sparser than $\boldsymbol{\beta}^*$ and it satisfies the constraint of problem (52), contradicting the optimality of $\boldsymbol{\beta}^*$.

Since $\|\mathbf{x}_i - \mathbf{X}\boldsymbol{\beta}^*\|_2 \leq 1$, $\|\mathbf{X}\boldsymbol{\beta}^*\|_2 \leq 2$. Also,

$$\sigma_{\min}(\mathbf{X}_{\boldsymbol{\beta}^*}^\top \mathbf{X}_{\boldsymbol{\beta}^*})\|\boldsymbol{\beta}^*\|_2^2 \leq \|\mathbf{X}\boldsymbol{\beta}^*\|_2^2 \leq 4,$$

it follows that $\|\boldsymbol{\beta}^*\|_2^2 \leq \frac{4}{\sigma_{\mathbf{X}}^{*2}}$. By Cauchy-Schwarz inequality, $\|\boldsymbol{\beta}^*\|_1 \leq \frac{2\sqrt{r^*}}{\sigma_{\mathbf{X}}^*}$ and $\|\mathbf{N}\boldsymbol{\beta}^*\|_2 \leq \|\boldsymbol{\beta}^*\|_1\delta \leq \frac{2\delta\sqrt{r^*}}{\sigma_{\mathbf{X}}^*}$. Therefore,

$$\begin{aligned}
&\|\mathbf{x}_i - \mathbf{Y}\boldsymbol{\beta}^*\|_2 = \|\mathbf{x}_i - \mathbf{X}\boldsymbol{\beta}^* + \mathbf{N}\boldsymbol{\beta}^*\|_2 \\
&\leq \|\mathbf{x}_i - \mathbf{X}\boldsymbol{\beta}^*\|_2 + \|\mathbf{N}\boldsymbol{\beta}^*\|_2 \leq c^* + \frac{2\delta\sqrt{r^*}}{\sigma_{\mathbf{X}}^*},
\end{aligned}$$

so that $\boldsymbol{\beta}^*$ is a feasible for problem (43). To prove that $\boldsymbol{\beta}^*$ is also an optimal solution to (43), suppose this is not the case, and an optimal solution to (43) is a vector $\boldsymbol{\beta}'$ such that $\|\mathbf{x}_i - \mathbf{Y}\boldsymbol{\beta}'\|_2 \leq c^* + \frac{2\delta\sqrt{r^*}}{\sigma_{\mathbf{X}}^*}$ and $\|\boldsymbol{\beta}'\|_0 = r < r^*$. $\mathbf{Y}_{\boldsymbol{\beta}'}$ is a full column rank matrix, otherwise a sparser solution can be obtained as vector whose support corresponds to the maximal linear independent set of columns of $\mathbf{Y}_{\boldsymbol{\beta}'}$. We have

$$d(\mathbf{x}_i, \mathbf{H}_{\mathbf{Y}_{\boldsymbol{\beta}'}}) \leq \|\mathbf{x}_i - \mathbf{Y}\boldsymbol{\beta}'\|_2 \leq c^* + \frac{2\delta\sqrt{r^*}}{\sigma_{\mathbf{X}}^*}.$$

According to Lemma B.1, we have

$$\begin{aligned}
&|d(\mathbf{x}_i, \mathbf{H}_{\mathbf{X}_{\boldsymbol{\beta}'}}) - d(\mathbf{x}_i, \mathbf{H}_{\mathbf{Y}_{\boldsymbol{\beta}'}})| \leq \frac{\sqrt{r}\delta}{\sigma_{\mathbf{Y},r} - \sqrt{r}\delta} \\
&= \frac{\delta}{\bar{\sigma}_{\mathbf{Y},r} - \delta} \leq \frac{\delta}{\bar{\sigma}_{\mathbf{Y}}^* - \delta} \\
&\Rightarrow d(\mathbf{x}_i, \mathbf{H}_{\mathbf{X}_{\boldsymbol{\beta}'}}) \leq c^* + \frac{2\delta\sqrt{r^*}}{\sigma_{\mathbf{X}}^*} + \frac{\delta}{\bar{\sigma}_{\mathbf{Y}}^* - \delta} = c^* + \tau_0.
\end{aligned}$$

However, according to the optimality of $\boldsymbol{\beta}^*$ in the noisy ℓ^0-SSC problem (3), we have

$$\begin{aligned}
&d(\mathbf{x}_i, \mathbf{H}_{\mathbf{X}_{\boldsymbol{\beta}'}}) - c^* = d(\mathbf{x}_i, \mathbf{H}_{\mathbf{X}_{\boldsymbol{\beta}'}}) - d(\mathbf{x}_i, \mathbf{H}_{\mathbf{X}_{\boldsymbol{\beta}^*}}) \\
&\geq (r^* - r)\lambda \geq \lambda > \tau_0
\end{aligned}$$

This contradiction shows that $\boldsymbol{\beta}^*$ is an optimal solution to (43). □

B.5 PROOF OF LEMMA B.3

Proof of Lemma B.3. (44) is equivalent to the following problem

$$\min_{\boldsymbol{\beta}} \|\boldsymbol{\beta}\|_0 \quad s.t.\ \mathbf{y} = \mathbf{Y}\boldsymbol{\beta},\ \|\mathbf{x}_i - \mathbf{y}\|_2 \leq c_0,\ \boldsymbol{\beta}_i = 0. \tag{53}$$

We show that the points (columns) of $\mathbf{Y}_{\boldsymbol{\beta}^*}$ must come from subspace $\mathcal{S}_k$. To see this, suppose some columns of $\mathbf{Y}_{\boldsymbol{\beta}^*}$ come from different subspaces. We first have $\|\boldsymbol{\beta}^*\|_0 \leq d_k$. To see this, we can choose some $\mathbf{y}' \in \mathcal{S}_k$ such that $\|\mathbf{y}' - \mathbf{x}_i\|_2 \leq c_0$ since $c_0 \geq d(\mathbf{x}_i, \mathcal{S}_k)$. Also, d_k points in $\mathbf{Y}^{(k)}$ can linearly represent $\mathbf{y}'$ since $\mathbf{Y}^{(k)}$ is in general position, and it follows that $\|\boldsymbol{\beta}^*\|_0 \leq d_k$ due to the optimality of $\boldsymbol{\beta}^*$.

Also, $\mathbf{Y}_{\boldsymbol{\beta}^*}$ has full column rank, so that subspace $\mathbf{H}_{\mathbf{Y}_{\boldsymbol{\beta}^*}} \in \mathcal{H}_{\mathbf{y}_i,d_k}$. Let $\mathbf{y}^* = \mathbf{Y}\boldsymbol{\beta}^*$, then $\mathbf{y}^* \in \mathbf{H}_{\mathbf{Y}_{\boldsymbol{\beta}^*}} \cap \mathbf{B}(\mathbf{x}_i, c_0)$ which contradicts the fact that $\mathbf{B}(\mathbf{x}_i, c_0) \cap \mathbf{H} = \emptyset$ for any $\mathbf{H} \in \mathcal{H}_{\mathbf{y}_i,d_k}$. Therefore, columns of $\mathbf{Y}_{\boldsymbol{\beta}^*}$ must come from $\mathcal{S}_k$. □

B.6 PROOF OF THEOREM A.4

Proof of Theorem A.4. This theorem can be proved by checking that the conditions in Theorem A.2 are satisfied. □

B.7 PROOF OF THEOREM A.7

In order to prove this theorem, the following lemma is presented and it provides the geometric concentration inequality for the distance between a point $\mathbf{y} \in \mathbf{Y}^{(k)}$ and any of its external subspaces. It renders a lower bound for M_i, namely the minimum distance between $\mathbf{y}_i \in \mathcal{S}_k$ and its external subspaces.

Lemma B.5. Under randomized models, given $1 \leq k \leq K$ and $\mathbf{y} \in \mathbf{Y}^{(k)}$, suppose $\mathbf{H} \in \mathcal{H}_{\mathbf{y}_i,d_k}$ is any external subspace of $\mathbf{y}$. Then for any $t > 0$,

$$\Pr[d(\mathbf{y}, \mathbf{H}) \geq 1 - 2t\sqrt{d_k - 1} - t^2] \geq 1 - 8\exp(-\frac{d_k t^2}{2}). \tag{54}$$

Proof of Lemma B.5. Let $\mathbf{H}$ be a fixed subspace of dimension $d_e \leq d_k$, and $\mathbf{y} \notin \mathbf{H}$. Since $\mathbf{y} \in \mathcal{S}_k$ and $\mathbf{y} \notin \mathbf{H}$. Let $\mathbf{U}_{\mathcal{S}_k} = \begin{bmatrix} \mathbf{I}_{d_k} \\ \mathbf{0} \end{bmatrix} \in \mathbb{R}^{d \times d_k}$ be the orthonormal basis of $\mathcal{S}_k$ under which the isotropic random vector $\mathbf{y}$ in $\mathcal{S}_k$ satisfies $\mathbb{E}[\mathbf{y}\mathbf{y}^\top] = \begin{bmatrix} \mathbf{I}_{d_k} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix}$. It follows that more columns vectors can be added to $\mathbf{U}_{\mathcal{S}_k}$ to form a orthonormal basis $\mathbf{U} \in \mathbb{R}^{d \times d'}$ for the minimum subspace that contains $\mathcal{S}_k$ and $\mathbf{H}$. It can be verified that $d_k + 1 \leq d' \leq \min\{d_k + d_e, d\}$ because $\mathbf{H} \neq \mathcal{S}_k$. Note that $\mathbf{U}$ can be represented as a block matrix as $\mathbf{U} = \begin{bmatrix} \mathbf{I}_{d_k} & \mathbf{0} \\ \mathbf{0} & \mathbf{U}' \end{bmatrix}$ where $\mathbf{U}' \in \mathbb{R}^{(d-d_k)\times(d'-d_k)}$ has orthonormal columns. It can be verified that the basis of $\mathbf{H}$ can be represented as $\mathbf{U}_{\mathbf{H}} = \begin{bmatrix} \mathbf{I}_{d_e-d'+d_k} & \mathbf{0} \\ \mathbf{0} & \mathbf{U}' \end{bmatrix}$. Note that if $d_e - d' + d_k = 0$, $\mathbf{U}_{\mathbf{H}} = \begin{bmatrix} \mathbf{0} \\ \mathbf{U}' \end{bmatrix}$. Then $\mathbb{P}_{\mathbf{H}}(\mathbf{y}) = \mathbf{U}_{\mathbf{H}}\mathbf{U}_{\mathbf{H}}^\top\mathbf{y}$, and we have

$$\begin{aligned}
&\mathbb{E}[\|\mathbb{P}_{\mathbf{H}}(\mathbf{y})\|_2^2] = \mathbb{E}[\mathbf{y}^\top\mathbf{U}_{\mathbf{H}}\mathbf{U}_{\mathbf{H}}^\top\mathbf{U}_{\mathbf{H}}\mathbf{U}_{\mathbf{H}}^\top\mathbf{y}] \\
&= \mathbb{E}[\mathrm{Tr}(\mathbf{y}^\top\mathbf{U}_{\mathbf{H}}\mathbf{U}_{\mathbf{H}}^\top\mathbf{y})] \\
&= \mathbb{E}[\mathrm{Tr}(\mathbf{U}_{\mathbf{H}}^\top\mathbf{y}\mathbf{y}^\top\mathbf{U}_{\mathbf{H}})] \\
&= \mathrm{Tr}(\mathbf{U}_{\mathbf{H}}^\top\mathbb{E}[\mathbf{y}\mathbf{y}^\top]\mathbf{U}_{\mathbf{H}}) \\
&= \mathrm{Tr}\Big(\begin{bmatrix} \mathbf{I}_{d_e-d'+d_k} & \mathbf{0} \\ \mathbf{0} & \mathbf{U}' \end{bmatrix}^\top \begin{bmatrix} \mathbf{I}_{d_k} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{I}_{d_e-d'+d_k} & \mathbf{0} \\ \mathbf{0} & \mathbf{U}' \end{bmatrix}\Big) \\
&= d_e - d' + d_k \leq d_e - 1 \leq d_k - 1
\end{aligned} \tag{55}$$

According to the concentration inequality in section 5.2 of (Aubrun & Szarek, 2017), for any $t > 0$,

$$\Pr[|\|\mathbb{P}_{\mathbf{H}}(\mathbf{y})\|_2 - \sqrt{d_e - d' + d_k}| \geq t] \leq 8\exp(-\frac{d_k t^2}{2}) \tag{56}$$

Now let $\mathbf{H}$ be spanned by data from $\mathbf{Y}$, i.e. $\mathbf{H} = \mathbf{H}_{\{\mathbf{y}_{i_j}\}_{j=1}^{d_e}}$, where $\{\mathbf{y}_{i_j}\}_{j=1}^{d_e}$ are any d_e linearly independent points that does not contain $\mathbf{y}$. For any fixed points $\{\mathbf{y}_{i_j}\}_{j=1}^{d_e}$, (56) holds. Let A be the event that $|\mathbb{P}_{\mathbf{H}}(\mathbf{y}) - \sqrt{d_e - d' + d_k}| \geq t$, we aim to integrate the indicator function $\mathbb{1}_A$ with respect to the random vectors, i.e. $\mathbf{y}$ and $\{\mathbf{y}_{i_j}\}_{j=1}^{d_e}$, to obtain the probability that A happens over these random vectors. Let $\mathbf{y} = \mathbf{y}_i$, using Fubini theorem, we have

$$\begin{aligned}
\Pr[A] &= \int_{\times_{j=1}^{n}\mathcal{S}^{(j)}} \mathbb{1}_A \otimes_{j=1}^{n} d\mu^{(j)} \\
&= \int_{\times_{j\neq i}\mathcal{S}^{(j)}} \Pr[A|\{\mathbf{y}_j\}_{j\neq i}] \otimes_{j\neq i} d\mu^{(j)} \\
&\leq \int_{\times_{j\neq i}\mathcal{S}^{(j)}} 8\exp(-\frac{d_k t^2}{2}) \otimes_{j\neq i} d\mu^{(j)} = 8\exp(-\frac{d_k t^2}{2})
\end{aligned} \tag{57}$$

where $\mathcal{S}^{(j)} \in \{\mathcal{S}_k\}_{k=1}^{K}$ is the subspace that $\mathbf{y}_j$ lies in, and $\mu^{(j)}$ is the probabilistic measure of the distribution in $\mathcal{S}^{(j)}$. The last inequality is due to (56).

Note that for any $\mathbf{y}$'s external subspace $\mathbf{H} = \mathbf{H}_{\{\mathbf{y}_{i_j}\}_{j=1}^{d_e}}$, $d(\mathbf{y}, \mathbf{H}) = \sqrt{\|\mathbf{y}\|_2^2 - \|\mathbb{P}_{\mathbf{H}}(\mathbf{y})\|_2^2} = \sqrt{d_k - \|\mathbb{P}_{\mathbf{H}}(\mathbf{y})\|_2^2}$. According to (57), we have

$$\Pr[d(\mathbf{y}, \mathbf{H}) \geq 1 - 2t\sqrt{d_k - 1} - t^2] \geq 1 - 8\exp(-\frac{d_k t^2}{2}). \tag{58}$$

□

Lemma B.6. Let $\mathbf{Y} \in \mathbb{R}^{d\times r}$ be a submatrix of $\mathbf{Y}^{(k)}$ with $\mathrm{rank}(\mathbf{Y}) = r$ and $r \leq d_k$. Then with probability at least $1-\exp(-d_k)$, $\sigma_{\min}(\mathbf{Y}_{\boldsymbol{\beta}}) \geq \frac{1}{\sqrt{d_{\max}}}\big(1-14\sqrt[3]{4}(\frac{M^2 d}{n})^{2/3}\big) = \sigma'_{\min}$ for any $\boldsymbol{\beta} \in \mathbb{R}^n$ such that $\|\boldsymbol{\beta}\|_0 = r$

Proof. Let $\mathbf{Y}' = \sqrt{d_k}\mathbf{Y}$. Let $\mathbf{U}^{(k)}$ be the orthogonal basis for $\mathcal{S}_k$, and $\mathbf{Y}'$ can be expressed by $\mathbf{Y}' = \mathbf{U}^{(}k))\mathbf{C}$, where $\mathbf{C} \in \mathbb{R}^{d_k\times r}$ and it can be verified that $\mathbf{C}^i$ is uniformly distributed on a sphere centered at the origin with radius $\sqrt{d_k}$, and $\mathbf{C}^i$ is a isotropic random vector for all $i \in [r]$. Let $M_k = \sup_{0\leq t<\sqrt{d_k}} \frac{2t^3 \arccos\frac{t}{\sqrt{d_k}}}{\pi}$, it can be verified that M_k satisfies the following condition: for any $t > 0$, any $i \in [n]$ and any vector $\mathbf{v}$ with unit ℓ^2-norm,

$$\Pr[|\langle \mathbf{C}^i, \mathbf{v}\rangle| > t] \leq \frac{M_k}{t^3}. \tag{59}$$

According to Theorem 2.1 in (Yaskov, 2015) and condition (59), with proability at least $1 - \exp(-d_k)$, $\sigma_{\min}(\mathbf{C}) \geq \frac{1}{\sqrt{d_{\max}}}\big(1 - 14\sqrt[3]{4}(\frac{M^2 d}{n})^{2/3}\big) = \sigma'_{\min}$ □

Proof of Theorem A.7. According to (Yaskov, 2014) and condition (a), with probability at least $1 - \exp(-d)$, $\sigma_{\min}(\mathbf{Y}_{\boldsymbol{\beta}} \geq \sqrt{196Md+1} - 14\sqrt{Md}$ for any $\boldsymbol{\beta} \in \mathbb{R}^n$ such that $\|\boldsymbol{\beta}\|_0 = r \leq d, \mathrm{rank}(\mathbf{Y}_{\boldsymbol{\beta}}) = \|\boldsymbol{\beta}\|_0$. It follows that $\sigma_{\mathbf{Y},r} \geq \sqrt{196Md+1} - 14\sqrt{Md}$. By Weyl (Weyl, 1912), $|\sigma_{\min}(\mathbf{X}_{\boldsymbol{\beta}}) - \sigma_{\min}(\mathbf{Y}_{\boldsymbol{\beta}})| \leq \|\mathbf{N}_{\boldsymbol{\beta}}\|_2 \leq \delta\sqrt{r_0}$. Therefore, $\sigma_{\min}(\mathbf{X}_{\boldsymbol{\beta}}) \geq \sqrt{196Md+1} - 14\sqrt{Md} - \delta\sqrt{r_0} > 0$ if $\delta < \frac{\sqrt{196Md+1}-14\sqrt{Md}}{\sqrt{r_0}} = c$. It can be verified that (36), (37) and (38) guarantee (29), (30) and (31) in Theorem A.4 respectively, therefore, the conclusion holds. □

B.8 PROOF OF THEOREM 3.1

Before presenting the proof, we introduce necessary lemmas.

It is proved that the low rank approximation $\bar{\mathbf{X}}$ is close to $\mathbf{X}$ in terms of the spectral norm (Halko et al., 2011):

Lemma B.7. (Corollary 10.9 in (Halko et al., 2011)) Let $p_0 \geq 2$ be an integer and $p' = p - p_0 \geq 4$, then with probability at least $1 - 6e^{-p}$, the spectral norm of $\mathbf{X} - \widehat{\mathbf{X}}$ is bounded by

$$\|\mathbf{X} - \widehat{\mathbf{X}}\|_2 \leq C_{p,p_0} \tag{60}$$

where

$$C_{p,p_0} \triangleq \big(1 + 17\sqrt{1 + \frac{p_0}{p'}}\big)\sigma_{p_0+1} + \frac{8\sqrt{p}}{p'+1}\big(\sum_{j>p_0}\sigma_j^2\big)^{\frac{1}{2}} \tag{61}$$

and $\sigma_1 \geq \sigma_2 \geq \ldots$ are the singular values of $\mathbf{X}$.

In addition, we have the following lemma on the perturbation bound for the distance between a data point and a subspace before and after the projection $\mathbf{P}$. Each subspace $\mathcal{S}_k$ is transformed into $\tilde{\mathcal{S}}_k = \mathbf{P}(\mathcal{S}_k)$ with dimension $\tilde{d}_k$.

Lemma B.8. Let $\boldsymbol{\beta} \in \mathbb{R}^n$, $\tilde{\mathbf{y}}_i = \mathbf{P}\mathbf{y}_i$, $\mathbf{H}_{\mathbf{Y}_{\boldsymbol{\beta}}}$ is an external subspace of $\mathbf{y}_i$, $\tilde{\mathbf{Y}}_{\boldsymbol{\beta}} = \mathbf{P}(\mathbf{Y}_{\boldsymbol{\beta}})$ and $\tilde{\mathbf{Y}}_{\boldsymbol{\beta}}$ has full column rank. Then

$$|d(\mathbf{y}_i, \mathbf{H}_{\mathbf{Y}_{\boldsymbol{\beta}}}) - d(\tilde{\mathbf{y}}_i, \mathbf{H}_{\tilde{\mathbf{Y}}_{\boldsymbol{\beta}}})| \leq C_{p,p_0}\big(1 + \frac{1}{\min_{1\leq r\leq \tilde{d}_k}\sigma_{\mathbf{Y},r} - C_{p,p_0} - 2\delta\sqrt{\tilde{d}_k}}\big) \tag{62}$$

for any $1 \leq i \leq n$ and $\mathbf{y}_i \in \mathcal{S}_k$.

Proof of Lemma B.8. This lemma can be proved by applying Lemma B.4. □

The following notations are defined before presenting the proof of Theorem 3.1.

$$\tilde{M}_i \triangleq \min\{d(\tilde{\mathbf{y}}_i, \mathbf{H})\colon \mathbf{H} \in \mathcal{H}_{\tilde{\mathbf{y}}_i, \tilde{d}_k}\}, \tag{63}$$

where $\mathcal{H}_{\tilde{\mathbf{y}}_i, \tilde{d}_k}$ is all the external subspace for $\tilde{\mathbf{y}}_i$ in the transformed space by $\mathbf{P}$.

$$\bar{\sigma}_{\tilde{\mathbf{Y}},r} \triangleq \min_{\boldsymbol{\beta}:\|\boldsymbol{\beta}\|_0=r, \mathrm{rank}(\tilde{\mathbf{Y}}_{\boldsymbol{\beta}})=\|\boldsymbol{\beta}\|_0} \sigma_{\min}(\tilde{\mathbf{Y}}_{\boldsymbol{\beta}}), \tag{64}$$

$$\sigma_{\tilde{\mathbf{X}},r} \triangleq \min\{\sigma_{\min}(\tilde{\mathbf{X}}_{\boldsymbol{\beta}})\colon 1 \leq \|\boldsymbol{\beta}\|_0 \leq r\}, \tag{65}$$

$$\tilde{\mu}_r \triangleq \frac{\delta}{\min_{1\leq r<r_0}\bar{\sigma}_{\mathbf{Y},r} - \delta}. \tag{66}$$

Proof of Theorem 3.1. For any matrix $\mathbf{A} \in \mathbb{R}^{p\times q}$, we first show that multiplying $\mathbf{Q}$ to the left of $\mathbf{A}$ would not change its spectrum. To see this, let the singular value decomposition of $\mathbf{A}$ be $\mathbf{A} = \mathbf{U}_\mathbf{A}\mathbf{\Sigma}\mathbf{V}_\mathbf{A}^\top$ where $\mathbf{U}_\mathbf{A}$ and $\mathbf{V}_\mathbf{A}$ have orthonormal columns with $\mathbf{U}_\mathbf{A}^\top\mathbf{U}_\mathbf{A} = \mathbf{V}_\mathbf{A}^\top\mathbf{V}_\mathbf{A} = \mathbf{I}$. Then $\mathbf{QA} = \mathbf{U}_{\mathbf{QA}}\mathbf{\Sigma}\mathbf{V}_{\mathbf{QA}}$ is the singular value decomposition of $\mathbf{QA}$ with $\mathbf{U}_{\mathbf{QA}} = \mathbf{QU}_\mathbf{A}$ and $\mathbf{V}_{\mathbf{QA}} = \mathbf{V}_\mathbf{A}$. This is because the columns of $\mathbf{U}_{\mathbf{QA}}$ are orthonormal since the columns $\mathbf{Q}$ are orthonormal: $\mathbf{U}_{\mathbf{QA}}^\top\mathbf{U}_{\mathbf{QA}} = \mathbf{U}_\mathbf{A}^\top\mathbf{Q}^\top\mathbf{QU}_\mathbf{A} = \mathbf{I}$, and $\mathbf{\Sigma}$ is a diagonal matrix with nonnegative diagonal elements. It follows that $\sigma_{\min}(\mathbf{QA}) = \sigma_{\min}(\mathbf{A})$ for any $\mathbf{A} \in \mathbb{R}^{p\times q}$.

For a point $\mathbf{x}_i = \mathbf{y}_i + \mathbf{n}_i$, after projection via $\mathbf{P}$, we have the projected noise $\tilde{\mathbf{n}}_i = \mathbf{P}\mathbf{n}_i$. Because

$$\|\tilde{\mathbf{n}}_i\|_2 = \|\mathbf{P}\mathbf{n}_i\|_2 = \|\mathbf{Q}^\top\mathbf{n}_i\|_2 \leq \|\mathbf{Q}\|_2\|\mathbf{n}_i\|_2 \leq \|\mathbf{n}_i\|_2 \leq \delta, \tag{67}$$

the magnitude of the noise in the projected data is also bounded by δ. Also,

$$\|\tilde{\mathbf{x}}_i\|_2 = \|\mathbf{Q}^\top \mathbf{x}_i\|_2 \leq \|\mathbf{x}_i\|_2 \leq 1, \tag{68}$$

Let $\boldsymbol{\beta} \in \mathbb{R}^n$, $\tilde{\mathbf{Y}}_{\boldsymbol{\beta}} = \mathbf{P}\mathbf{Y}_{\boldsymbol{\beta}}$ with $\|\boldsymbol{\beta}\|_0 = r$. Then $\sigma_{\min}(\mathbf{Q}\tilde{\mathbf{Y}}_{\boldsymbol{\beta}}) = \sigma_{\min}(\tilde{\mathbf{Y}}_{\boldsymbol{\beta}}))$. Since

$$\begin{aligned}
&|\sigma_{\min}(\tilde{\mathbf{Y}}_{\boldsymbol{\beta}}) - \sigma_{\min}(\mathbf{Y}_{\boldsymbol{\beta}})| = |\sigma_{\min}(\mathbf{Q}\tilde{\mathbf{Y}}_{\boldsymbol{\beta}}) - \sigma_{\min}(\mathbf{Y}_{\boldsymbol{\beta}})| \\
&\leq \|\mathbf{Q}\tilde{\mathbf{Y}}_{\boldsymbol{\beta}} - \mathbf{Y}_{\boldsymbol{\beta}}\|_2 \\
&= \|\mathbf{Q}\mathbf{Q}^\top \mathbf{Y}_{\boldsymbol{\beta}} - \mathbf{Y}_{\boldsymbol{\beta}}\|_2 \\
&= \|\mathbf{Q}\mathbf{Q}^\top \mathbf{X}_{\boldsymbol{\beta}} - \mathbf{X}_{\boldsymbol{\beta}} + \mathbf{N}_{\boldsymbol{\beta}} - \mathbf{Q}\mathbf{Q}^\top \mathbf{N}_{\boldsymbol{\beta}}\|_2 \\
&\leq C_{p,p_0} + \|\mathbf{N}_{\boldsymbol{\beta}}\|_F + \|\mathbf{Q}\mathbf{Q}^\top \mathbf{N}_{\boldsymbol{\beta}}\|_F \\
&\leq C_{p,p_0} + 2\delta\sqrt{r}
\end{aligned} \tag{69}$$

Therefore, it follows from (69) that if

$$C_{p,p_0} + 2\delta\sqrt{\tilde{d}_{\max}} < \min_{k=1,\ldots,K} \sigma_{\mathbf{Y}}^{(k)}, \tag{70}$$

then $\tilde{\mathbf{Y}}$ is also in general position.

In addition, since $\lambda \geq \frac{1}{r_0}$, we have $\lambda\|\tilde{\boldsymbol{\beta}}^*\|_0 \leq L(\mathbf{0}) \leq 1$, and it follows that $\|\tilde{\boldsymbol{\beta}}^*\|_0 \leq \frac{1}{\lambda} \leq r_0$.

Based on (69) we have

$$|\bar{\sigma}_{\tilde{\mathbf{Y}},r} - \bar{\sigma}_{\mathbf{Y},r}| \leq C_{p,p_0} + 2\delta\sqrt{r_0}, \tag{71}$$

it follows that $\delta < \min_{1\leq r<r_0} \bar{\sigma}_{\tilde{\mathbf{Y}},r}$ because $\delta < \min_{1\leq r<r_0} \bar{\sigma}_{\mathbf{Y},r} - C_{p,p_0} - 2\delta\sqrt{r_0}$.

Again, for $\boldsymbol{\beta} \in \mathbb{R}^n$ with $\|\boldsymbol{\beta}\|_0 = r \leq r_0$, we have

$$\begin{aligned}
&|\sigma_{\min}(\tilde{\mathbf{X}}_{\boldsymbol{\beta}}) - \sigma_{\min}(\mathbf{X}_{\boldsymbol{\beta}})| = |\sigma_{\min}(\mathbf{Q}\tilde{\mathbf{X}}_{\boldsymbol{\beta}}) - \sigma_{\min}(\mathbf{X}_{\boldsymbol{\beta}})| \\
&\leq \|\mathbf{Q}\tilde{\mathbf{X}}_{\boldsymbol{\beta}} - \mathbf{X}_{\boldsymbol{\beta}}\|_2 \\
&= \|\mathbf{Q}\mathbf{Q}^\top \mathbf{X}_{\boldsymbol{\beta}} - \mathbf{X}_{\boldsymbol{\beta}}\|_2 = \|\widehat{\mathbf{X}} - \mathbf{X}_{\boldsymbol{\beta}}\|_2 \\
&\leq C_{p,p_0}
\end{aligned} \tag{72}$$

It can be verified that

$$|\sigma_{\tilde{\mathbf{X}},r} - \sigma_{\mathbf{X},r}| \leq C_{p,p_0} \tag{73}$$

Combining (73) and Lemma B.8, noting that $\sigma_{\mathbf{X},r_0} - C_{p,p_0}$, since

$$\begin{aligned}
&M_i - C_{p,p_0}(1 + \frac{1}{\min_{1\leq r\leq \tilde{d}_k} \sigma_{\mathbf{Y},r} - C_{p,p_0} - 2\delta\sqrt{\tilde{d}_k}}) \\
&> \delta + \frac{2\delta}{\sigma_{\mathbf{X},r_0} - C_{p,p_0}},
\end{aligned} \tag{74}$$

we have

$$\tilde{M}_{i,\delta} \triangleq \tilde{M}_i - \delta > \frac{2\delta}{\tilde{\sigma}_{\tilde{\mathbf{X}},r_0}}, \tag{75}$$

where $\mathbf{y}_i \in \mathcal{S}_k$.

Based on (71) and (73), we have

$$\tilde{\mu}_{r_0} < 1 - \frac{2\delta}{\sigma_{\tilde{\mathbf{X}},r_0}}, \tag{76}$$

because

$$\frac{\delta}{\min_{1\leq r<r_0} \bar{\sigma}_{\mathbf{Y},r_0} - C_{p,p_0} - 2\delta\sqrt{r_0} - \delta} < 1 - \frac{2\delta}{\sigma_{\mathbf{X},r_0} - C_{p,p_0}} \tag{77}$$

□

B.9 PROOF OF THEOREM 3.2

Proof of Theorem 3.2. It can be verified that $\tilde{M}_i \geq \frac{M_i}{1+\varepsilon}$. Let $\boldsymbol{\beta} \in \mathbb{R}^n$, $\tilde{\mathbf{Y}}_{\boldsymbol{\beta}} = \mathbf{P}\mathbf{Y}_{\boldsymbol{\beta}}$ with $\|\boldsymbol{\beta}\|_0 = r$ and $\operatorname{rank}(\mathbf{Y}_{\boldsymbol{\beta}}) = r$, then for any $\mathbf{u} \in \mathbb{R}^r$, $\|\tilde{\mathbf{Y}}_{\boldsymbol{\beta}}\mathbf{u}\|_2 = \|\mathbf{P}\mathbf{Y}_{\boldsymbol{\beta}}\mathbf{u}\|_2 \geq (1-\varepsilon)\|\mathbf{Y}_{\boldsymbol{\beta}}\mathbf{u}\|_2 \geq (1-\varepsilon)\sigma_{\min}(\mathbf{Y}_{\boldsymbol{\beta}})\|\mathbf{u}\|_2$. It follows that $\sigma_{\min}(\tilde{\mathbf{Y}}_{\boldsymbol{\beta}}) \geq (1-\varepsilon)\sigma_{\min}(\mathbf{Y}_{\boldsymbol{\beta}})$, and $\bar{\sigma}_{\tilde{\mathbf{Y}},r} \geq (1-\varepsilon)\bar{\sigma}_{\mathbf{Y},r}$. Similarly, $\sigma_{\min}(\tilde{\mathbf{X}}_{\boldsymbol{\beta}}) \geq (1-\varepsilon)\sigma_{\min}(\mathbf{X}_{\boldsymbol{\beta}})$ for $\boldsymbol{\beta} \in \mathbb{R}^n$, $\|\boldsymbol{\beta}\|_0 = r$ and $\operatorname{rank}(\mathbf{X}_{\boldsymbol{\beta}}) = r$. It follows that $\sigma_{\tilde{\mathbf{X}},r} \geq (1-\varepsilon)\sigma_{\mathbf{X},r}$. Since (16)-(20) hold, the conditions (29)-(33) required by Theorem A.4 on the projected data ($\tilde{\mathbf{Y}}$ and $\tilde{\mathbf{X}}$) also hold. Therefore, the subspace detection property holds with $\tilde{\boldsymbol{\beta}}^*$ for $\tilde{\mathbf{x}}_i$ with probability at least $1 - K\delta$ by the union bound when $p \geq \frac{d^2+d}{\delta(2\varepsilon-\varepsilon^2)^2}$. □

C BOUND FOR SUBOPTIMAL AND GLOBALLY OPTIMAL SOLUTIONS FOR NOISY ℓ^0-SSC AND NOISY-DR-ℓ^0-SSC

We further present the bound for the gap between $\widehat{\boldsymbol{\beta}}$ and $\boldsymbol{\beta}^*$, $\|\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}^*\|_2$, based on Theorem 5 in (Yang & Yu, 2019). Let $g(\boldsymbol{\beta}) = \|\mathbf{x}_i - \mathbf{X}\boldsymbol{\beta}\|_2^2$ and $\boldsymbol{\beta}^*$ be the globally optimal solution to (3), $\mathbf{S}^* = \operatorname{supp}(\boldsymbol{\beta}^*)$, $\widehat{\boldsymbol{\beta}}$ be the suboptimal solution to (3) obtained by PGD, $\widehat{\mathbf{S}} = \operatorname{supp}(\widehat{\boldsymbol{\beta}})$. The following theorem presents the bound for $\|\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}^*\|_2$.

Theorem C.1. (Theorem 5 in (Yang & Yu, 2019)) Suppose $\mathbf{X}_{\mathbf{S}\cup\mathbf{S}^*}$ has full column rank with $\kappa_0 \triangleq \sigma_{\min}(\mathbf{X}_{\mathbf{S}\cup\mathbf{S}^*}) > 0$ where $\mathbf{S}$ is the support of the initialization for PGD on problem (3). Let $\kappa > 0$ such that $2\kappa_0^2 > \kappa$ and b is chosen according to (78) as below:

$$0 < b < \min\{\min_{j\in\widehat{\mathbf{S}}} |\widehat{\boldsymbol{\beta}}_j|, \frac{\lambda}{\max_{j\notin\widehat{\mathbf{S}}} |\frac{\partial g}{\partial \boldsymbol{\beta}_j}|_{\boldsymbol{\beta}=\widehat{\boldsymbol{\beta}}}|}, \min_{j\in\mathbf{S}^*} |\boldsymbol{\beta}_j^*|, \frac{\lambda}{\max_{j\notin\mathbf{S}^*} |\frac{\partial g}{\partial \boldsymbol{\beta}_j}|_{\boldsymbol{\beta}=\boldsymbol{\beta}^*}|}\}. \tag{78}$$

Let $\mathbf{F} = (\widehat{\mathbf{S}} \setminus \mathbf{S}^*) \cup (\mathbf{S}^* \setminus \widehat{\mathbf{S}})$ be the symmetric difference between $\widehat{\mathbf{S}}$ and $\mathbf{S}^*$, then

$$\|\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}^*\|_2 \leq \frac{1}{2\kappa_0^2 - \kappa}\Big(\sum_{j\in\mathbf{F}\cap\widehat{\mathbf{S}}} (\max\{0, \frac{\lambda}{b} - \kappa|\widehat{\boldsymbol{\beta}}_j - b|\})^2 + \sum_{j\in\mathbf{F}\setminus\widehat{\mathbf{S}}} (\max\{0, \frac{\lambda}{b} - \kappa b\})^2\Big)^{\frac{1}{2}}. \tag{79}$$

Remark C.2. It is observed that the gap $\|\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}^*\|_2$ is small when $\frac{\lambda}{b} - \kappa|\widehat{\boldsymbol{\beta}}_j - b|$ for $j \in \mathbf{F}\cap\widehat{\mathbf{S}}$ and $\frac{\lambda}{b} - \kappa b$ are small. Based on this observation, Theorem C.3 establishes the conditions under which $\widehat{\boldsymbol{\beta}}$ is also an optimal solution to (3), i.e. $\widehat{\boldsymbol{\beta}} = \boldsymbol{\beta}^*$.

Define $\mathbf{S}^* = \operatorname{supp}(\boldsymbol{\beta}^*)$, $H^* = \max_{1\leq j\leq n} \operatorname{dist}(\boldsymbol{\beta}, \mathbf{H}_{\mathbf{X}_{\mathbf{S}^*\setminus\{j\}}})$, $\mu = \max\{H^* + \|\boldsymbol{\beta}_i - \mathbf{X}\boldsymbol{\beta}^*\|_2, 2\|\mathbf{x}_i - \mathbf{X}\widehat{\boldsymbol{\beta}}\|_2, 2\|\mathbf{x}_i - \mathbf{X}\boldsymbol{\beta}^*\|_2\}$, $\kappa_0 = \sigma_{\min}(\mathbf{X}_{\mathbf{S}\cup\mathbf{S}^*}) > 0$ where $\mathbf{S} = \operatorname{supp}(\boldsymbol{\beta}^{(0)})$. The following theorem demonstrates that $\widehat{\boldsymbol{\beta}} = \boldsymbol{\beta}^*$ if λ is two-side bounded and $\widehat{\boldsymbol{\beta}}_{\min} = \min_{t:\widehat{\boldsymbol{\beta}}_t\neq 0} |\widehat{\boldsymbol{\beta}}_t|$ is sufficiently large.

Theorem C.3. (Conditions that the suboptimal solution by PGD is also globally optimal) If

$$\widehat{\boldsymbol{\beta}}_{\min} \geq \frac{\mu}{\kappa_0^2} \tag{80}$$

and

$$\frac{\mu^2}{2\kappa_0^2} \leq \lambda \leq (\widehat{\boldsymbol{\beta}}_{\min} - \frac{\mu}{2\kappa_0^2})\mu, \tag{81}$$

then $\widehat{\boldsymbol{\beta}} = \boldsymbol{\beta}^*$.

Sketch of Proof. It can be verified that $\max\{0, \frac{\lambda}{b} - \kappa|\widehat{\boldsymbol{\beta}}_j - b|\} = 0$ and $\max\{0, \frac{\lambda}{b} - \kappa b\} = 0$ under the conditions (80) and (81), therefore, $\widehat{\boldsymbol{\beta}} = \boldsymbol{\beta}^*$ by applying Theorem C.1. □

D MORE DETAILS ABOUT EXPERIMENTS

D.1 DATASETS

We conduct experiments on the following datasets. COIL-20 dataset has 1440 images of size 32×32 for 20 objects with background removed in all images. The COIL-100 dataset contains 100 objects

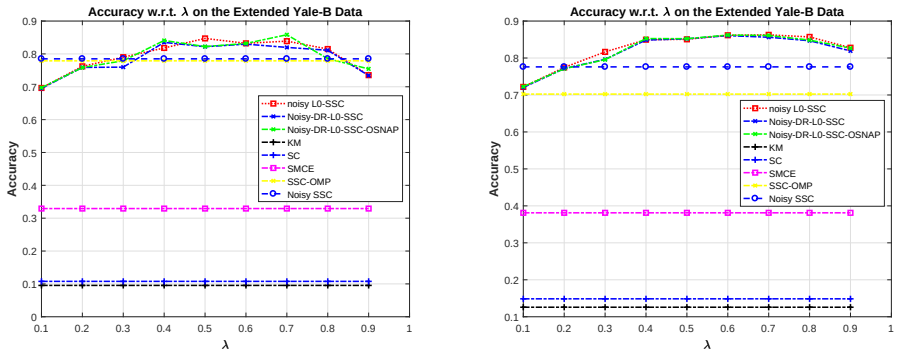
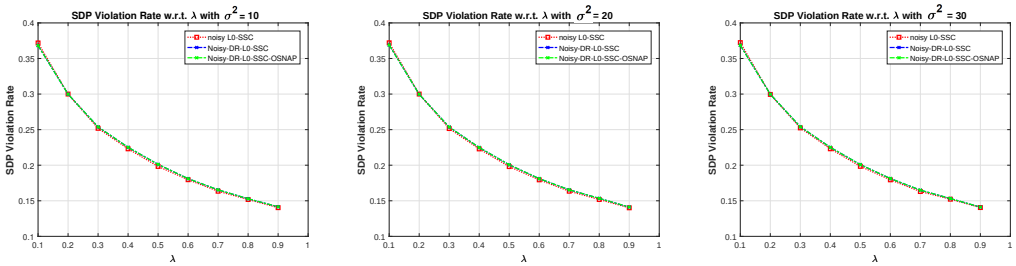


Figure 3: Accuracy (left) and NMI (right) with respect to different values of λ on the Extended Yale-B data set



(a) $\sigma^2 = 10$ (b) $\sigma^2 = 20$ (c) $\sigma^2 = 30$

(d) $\sigma^2 = 40$ (e) $\sigma^2 = 50$ (f) $\sigma^2 = 60$

Figure 4: The SDP violation rate with respect to λ for noisy ℓ^0-SSC, Noisy-DR-ℓ^0-SSC and Noisy-DR-ℓ^0-SSC-OSNAP with different noise levels σ^2. The SDP violation rates of Noisy-DR-ℓ^0-SSC and that for Noisy-DR-ℓ^0-SSC-OSNAP are almost the same, so their curves largely overlap each other.

with 72 images of size 32×32 for each object. The Extended Yale Face Database B (Yale-B) dataset contains face images for 38 subjects with 64 frontal face images taken under different illuminations for each subject. CMU Multi-PIE (MPIE) data (Gross et al., 2010) contains 8916 facial images captured in four sessions, namely S1-S4. The MNIST handwritten digits database has a total number of 70000 samples of dimensionality 1024 for digits from 0 to 9. The digits are normalized and centered in a fixed-size image.

D.2 More Experimental Results

Figure 3 illustrates how the accuracy and NMI vary with respect to λ on the Extended Yale-B (Yale-B) data set.

Figure 4(a) to Figure 4(f) illustrate SDP violation with respect to λ for different noise levels, justifying our theoretical finding that relatively large λ tends to preserve the subspace detection property

for noisy ℓ^0-SSC, Noisy-DR-ℓ^0-SSC and Noisy-DR-ℓ^0-OSNAP. It can be observed that larger λ consistenly renders lower SDP violation rate.